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# Automated Internal Defect Identification and Localization Based on a Near-Field SAR Millimeter-Wave Imaging System

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**ABSTRACT** Fast and cost-effective detection of internal defects is essential for structural integrity inspection in various applications such as manufacturing, construction, and aerospace. Current internal non-destructive testing (NDT) methods, such as computed tomography, can be costly, time-consuming, and constrained by object size. The lightweight and affordable millimeter-wave (mmWave) radar has demonstrated the capability of detecting objects beneath surfaces or obstacles through generated near-field synthetic aperture radar (SAR) images. However, its use in precise internal inspection has not been fully explored due to the significant noise in the SAR images and high rate of false identification. To enable accurate and fast inspections using the mmWave radar, this work establishes a robust and automated internal defect detection system. It employs a compact mmWave radar system mounted on a stable rail-based scanning mechanism, generating high-resolution near-field SAR images with an enhanced signal-to-noise ratio through denoising. For fast and accurate detection, an automated defect localization algorithm is developed. The accuracy of detection is ensured by modeling and separating internal defects from disturbances introduced by the scanning mechanism and noise. Experiments were conducted using 3D-printed blocks with synthetic defects to demonstrate the detection capability of the proposed system. Internal defects were accurately detected across variations in shape, material permittivity, depth, and size. The proposed method achieved an average accuracy of 91.7%, outperforming existing methods. The compact design of the radar system enables seamless integration with larger scanning systems, while the automated detection algorithm can be readily implemented as a software module within existing sensing systems. This integration of hardware and software components yields a versatile, low-cost framework for rapid internal health inspection that adapts to various industrial applications and object sizes.

**INDEX TERMS** Internal defect detection, near-field synthetic aperture radar (SAR) imaging, millimeter-wave (mmWave) radar, non-destructive testing (NDT), defect identification and localization.

## I. INTRODUCTION

Internal defect detection facilitates assessment of structural health, which is crucial for quality assurance in manufacturing, construction, and aircraft maintenance [1], [2]. Non-destructive testing (NDT) has been routinely utilized

for internal defect detection. It facilitates timely maintenance and prevents costly breakdowns or accidents. Current NDT methods for internal inspection employ various types of waves or energy sources to penetrate solid materials and analyze the reflected signals. They often require advanced and expensive equipment to ensure accurate and reliable results. For example, ultrasonic testing requires a coupling agent on the object's surface to ensure effective transmission

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of ultrasound waves [3]. This requirement increases operational costs and slows down the inspection process. Other NDT methods utilize higher-frequency waves with better penetration capabilities without requiring a coupling agent. For example, radiography testing (RT) uses high-ionizing radiation, such as X-rays. RT necessitates access restrictions and regulatory compliance, making installation and operation costly [4]. X-ray computed tomography (CT) is widely used for internal NDT as it produces highly detailed 3D images [5]. However, CT demands stringent safety measures and relies on a compact, complex scanning system. The scanning is typically conducted using a rotary scanner and motorized table, which increases the cost of operations and limits the sizes of inspected objects. Moreover, these X-ray images require exposure time, delaying result evaluation [6]. To enable fast and adaptive detection, Terahertz (THz) testing utilizes electromagnetic waves in the frequency range of 0.1 THz to 10 THz to penetrate non-metallic materials. Since electromagnetic waves are non-ionizing, using THz eliminates the need for strict safety protocols. However, it faces challenges when inspecting thicker components due to limited penetration capabilities [7], and high installation costs arising from expensive THz sources [8].

To ensure both penetration capability and cost-effectiveness, mmWave NDT systems have been applied for internal defect detection [9], [10], [11]. Similar to THz technology, mmWave systems utilize electromagnetic waves with a frequency range of 30–300 GHz, corresponding to a wavelength between 1 mm and 10 mm. Its wavelength allows signals to penetrate a wide range of dielectric materials such as composites, plastic, wood, concrete, ceramics, and clothing [12], [13]. mmWave systems are significantly cheaper and more readily available than THz systems due to using lower-frequency components [14]. However, current mmWave systems typically rely on numerous discrete components, such as filter modules, transmit source modules, and frequency multiplier modules, among others [15], [16], [17], [18]. In particular, necessary components like a vector network analyzer (VNA) [17] are typically large and heavy. The current complexity and size of mmWave systems make them difficult to move, limiting their ability to detect large-scale objects.

To enhance the mobility of mmWave radar systems, monolithic mmWave integrated circuits (MMICs) are considered due to their compact and integrated design [19], [20], [21]. MMICs integrate only key components such as transmitters, receivers, and amplifiers into a single, lightweight chip. These lightweight mmWave radars have been utilized, for example, in vehicles for advanced driver-assistance systems. They enable the detection of obstacles and passengers, even under challenging weather conditions such as fog, rain, and snow [22], [23]. For internal NDT applications, these lightweight and compact radars can be mounted on rails or drones for fast and on-the-spot large-scale scanning. It can reduce cost associated with transportation and the

construction of larger machines to accommodate different shapes.

For on-the-spot detection, the signals received from the sensor have to be processed real-time. Typically, the signals are processed using the synthetic aperture radar (SAR) technique to generate images [24]. The SAR technique works by transmitting radar signals, capturing their echoes from a target, and using the motion of the radar to synthesize a large virtual aperture for target image generation. SAR imaging is typically conducted in the far-field for surface surveillance, where the target is far enough from the radar system for the spherical wavefront to be approximated as planar [25]. However, for internal NDT, a near-field SAR approach is preferred since the sensor is generally closer to the objects for higher signal strength [26]. Furthermore, in near-field SAR, proximity to the target captures evanescent waves, enabling sub-wavelength resolution that is significantly decayed in far-field scenarios [27]. These benefits provide enhanced quality for imaging scenarios requiring detailed detection of internal structures.

However, using the near-field SAR MMIC-based mmWave system for defect detection can be time-consuming, as defects must be manually identified from the SAR images. For example, in [28], internal cracks in the rebar of reinforced concrete are manually identified from the generated SAR images based on the intensity differences between the cracks and the intact areas. Defects in various materials have been examined in a similar manual way using the near-field SAR MMIC-based mmWave system, including a 3D-printed sample made of polylactic acid (PLA) [29], metal plates, and carbon fiber reinforced plastic (CFRP) boards [30]. Furthermore, the generated SAR images often contain significant noise, which can obscure details and make manual defect detection challenging. The primary noise is speckle noise, caused by the constant interference of high-frequency radar waves with target areas [31]. This noise becomes particularly severe when the sensor is close to the object. The close distance leads to mutual interference and coupling signals within the internal SAR system [32], [33]. It also increases multi-path interference from the external environment [34], [35] and noise from vibrations during system scanning [36]. These noise sources reduce the contrast between defective and defect-free regions, making it difficult to identify defective areas and resulting in a high rate of false identification during manual detection.

Unknown defects may be detected, necessitating expert experience for accurate identification. To address these challenges, an automated and accurate defect detection system is needed to improve the reliability and efficiency of near-field SAR-based inspection systems. Although deep learning models have been trained to classify pixels as nominal or defective [37], [38], [39], these training-based methods heavily rely on the quantity and quality of the training data. And these models cannot be directly applied to inspecting new objects [40]. The presence of noise in the

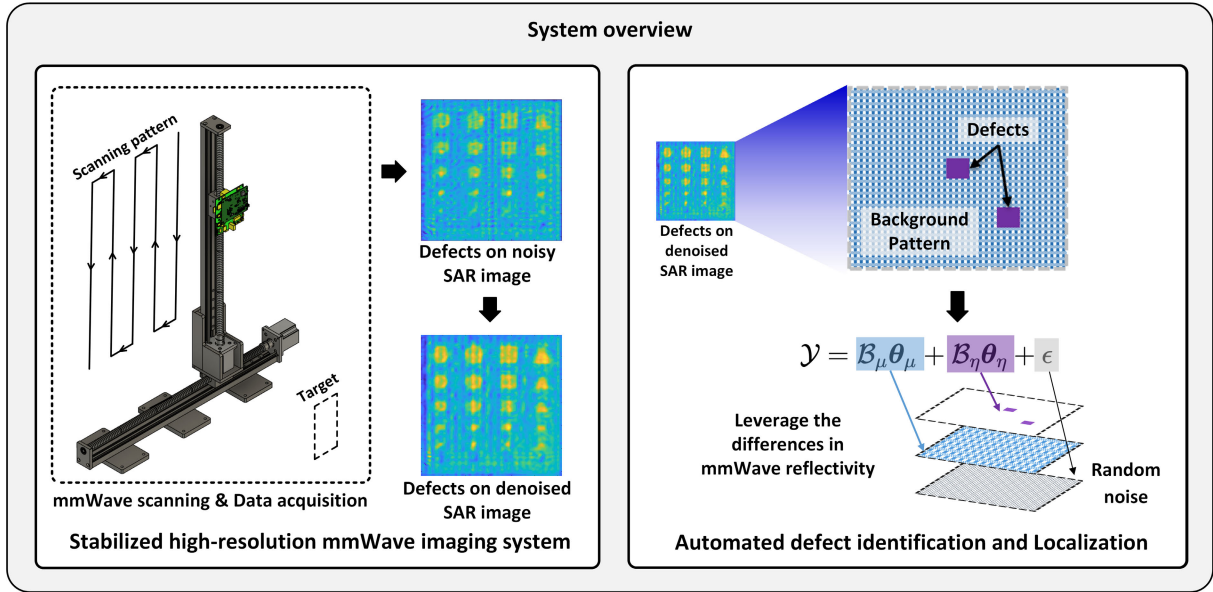


FIGURE 1. Overview of a complete mmWave imaging system designed for automated internal defect identification and localization.

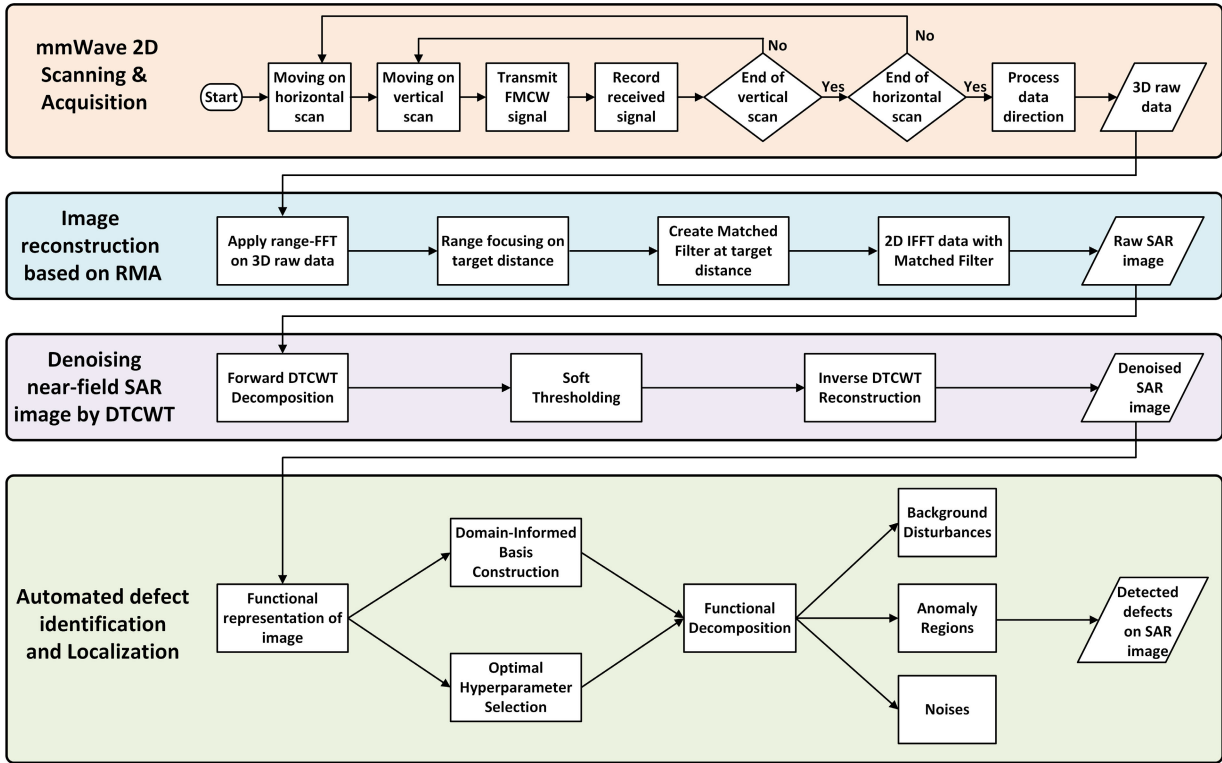


FIGURE 2. Flowchart of the overall algorithm framework.

near-field SAR images can also compromise the quality of the training data, while background disturbances induced by the scanning processes further affect detection and labeling accuracy [41]. Therefore, to accelerate the detection process, it is essential to develop an automated defect detection algorithm that is robust against noise and disturbances.

To enable fast and cost-effective internal NDT, this study focuses on developing a lightweight and automated system for internal defect detection and localization. The proposed system consists of: (1) a fast and affordable mmWave system that generates high-resolution, low-noise near-field SAR images, and (2) an automated defect detection algorithm

TABLE 1. List of abbreviations.

| Abbreviation | Definition                                   |
|--------------|----------------------------------------------|
| NDT          | non-destructive testing                      |
| mmWave       | millimeter wave                              |
| SAR          | synthetic aperture radar                     |
| VNA          | vector network analyzer                      |
| MMICs        | monolithic mmWave integrated circuits        |
| FMCW         | frequency-modulated continuous wave          |
| SISO         | single-input single-output antenna           |
| IF           | intermediate frequency                       |
| DTCWT        | dual-tree complex wavelet transform          |
| FMCW         | frequency-modulated continuous wave          |
| RMA          | range migration algorithm                    |
| PSNR         | peak signal-to-noise ratio                   |
| SSD          | smooth-sparse decomposition                  |
| PISSD        | process-informed smooth-sparse decomposition |
| MUT          | material under test                          |
| TP           | true positives                               |
| TN           | true negatives                               |
| FP           | false positives                              |
| FN           | false negatives                              |
| FPR          | false positive rate                          |
| FNR          | false negative rate                          |

that adapts to noises and disturbances to expedite the detection process. The proposed near-field SAR mmWave system allows for quicker scanning, covering inspection areas more rapidly than bulky systems. Additionally, the automated algorithm significantly reduces manual inspection time, offering a more efficient method for identifying defects. An illustration of the proposed system is shown in Fig. 1 and a flowchart of the overall algorithm framework is presented in Fig. 2.

The main contributions can be summarized as follows:

- A compact mmWave imaging system has been developed to employ a fast, low-cost mmWave radar sensor, a high-speed data acquisition module and a stabilized scanning mechanism.
- A denoising algorithm based on the dual-tree complex wavelet transform (DTCWT) has been validated in SAR image in near-field context, enhancing the robustness of mmWave imaging system.
- An automated internal defect identification and localization method has been proposed to address the challenges posed by non-smooth and dynamic background patterns in near-field SAR images.

The rest of the paper is organized as follows: Section II describes the design and development of the compact high-resolution mmWave imaging system for near-field SAR image generation. Section III introduces the proposed automated internal defect algorithm using near-field SAR images. The experiments, result discussion are presented in section IV. Section V discusses the limitations and challenges encountered during the study, along with potential directions for future research. Finally, the conclusion is summarized in section VI. Table 1 and Table 2 show the list of abbreviations and list of symbols, which help to improve the readability and accessibility of this paper.

TABLE 2. List of symbols.

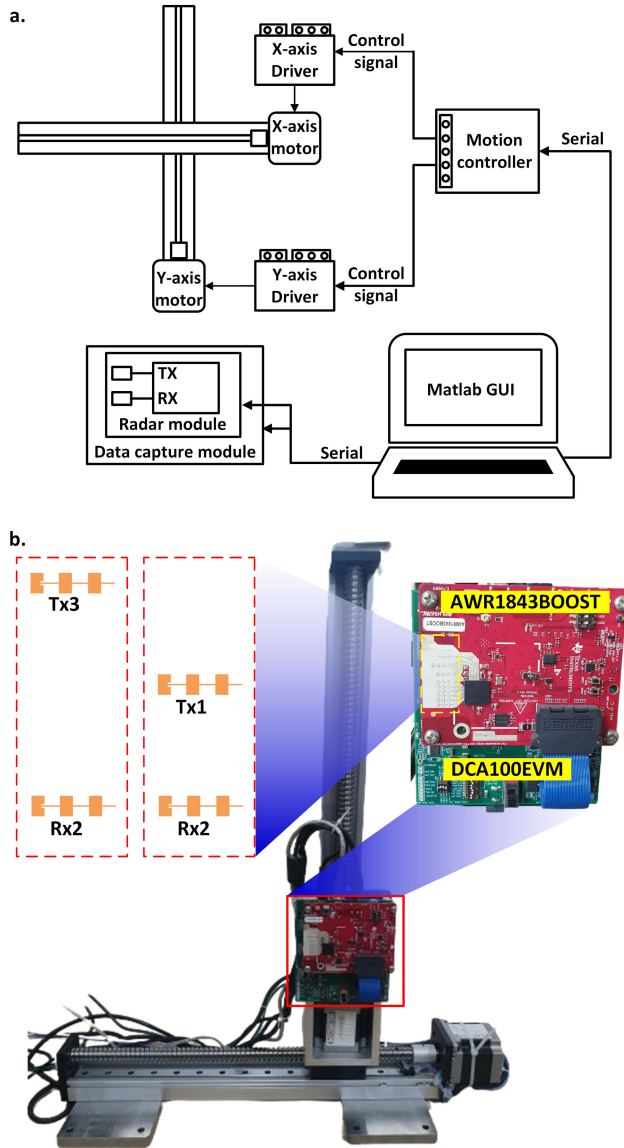
| Symbol                                 | Definition                                                                                 |
|----------------------------------------|--------------------------------------------------------------------------------------------|
| $R$                                    | distance between the radar and target aperture                                             |
| $c$                                    | speed of light                                                                             |
| $f_0$                                  | carrier frequency of the FMCW signal                                                       |
| $f_T$                                  | maximum swept frequency of the FMCW signal                                                 |
| $z_0$                                  | distance between scanning aperture of radar sensor and MUT                                 |
| $d_x$                                  | scanning step sizes following horizontal side                                              |
| $d_y$                                  | scanning step sizes following vertical side                                                |
| $D_x, D_y$                             | scanning aperture sizes following horizontal and vertical direction                        |
| $\delta_x, \delta_y$                   | cross-range resolution for near-field SAR imagery                                          |
| $\delta_z$                             | range resolution for near-field SAR imagery                                                |
| $\lambda_c$                            | center wavelength                                                                          |
| $N_s$                                  | number of samples for each chirp signal.                                                   |
| $H_p, V_p$                             | number of measurement points in the horizontal and vertical direction                      |
| $M$                                    | formation of the data cube                                                                 |
| $\Gamma$                               | RMA transform data in the cube using range-FFT                                             |
| $S$                                    | range focusing to determine the target planar                                              |
| $MF$                                   | matched filter                                                                             |
| $\hat{M}$                              | reconstructed near-field SAR imaging                                                       |
| $h_0(n), h_1(n)$                       | two filter of real part in forward DTCWT decomposition                                     |
| $g_0(n), g_1(n)$                       | two filter of imaginary part in forward DTCWT decomposition                                |
| $\overline{h_0(n)}, \overline{h_1(n)}$ | two filter of real part in inverse DTCWT reconstruction                                    |
| $\overline{g_0(n)}, \overline{g_1(n)}$ | two filter of imaginary part in inverse DTCWT reconstruction                               |
| $\mathcal{Y}$                          | image for decomposition                                                                    |
| $\mathcal{B}_\mu, \mathcal{B}_\eta$    | Tensor product of 1D bases for the background pattern and defect pattern                   |
| $\mu, \eta, \epsilon$                  | functions of background disturbances, defect regions, and random noises                    |
| $\theta_\mu, \theta_\eta$              | coefficients of $\mathcal{B}_\mu, \mathcal{B}_\eta$                                        |
| $N_{Bx}, N_{By}$                       | number of bases along the x and y axes to represent the pattern of background disturbances |
| $N_{Ax}, N_{Ay}$                       | number of bases along the x and y axes to identify the defects                             |

## II. COMPACT HIGH-RESOLUTION mmWAVE IMAGING SYSTEM

### A. STABILIZED SCANNING MECHANISM

In this section, the stabilized scanning mechanism to generate high resolution images is introduced. The block diagram of the established mmWave scanning system is shown in Fig. 3a. It contains a TI AWR1483BOOST for mmWave transceiver and a DCA1000 module for high-speed data acquisition. The radar module is based on a single-chip AWR1843 mmWave sensor, which integrates four receiving antennas and three transmitting antennas. The radar sensor supports 4 GHz available bandwidth with a frequency-modulated continuous wave (FMCW) signal from 77 GHz to 81 GHz. This radar sensor is interfaced with the DCA1000EVM data capture module for high-speed data acquisition to the laptop over an Ethernet connection. To improve the stability, the compact defect sensing system containing the radar and data capture module is mounted on two slide rails. The sensor is moved continuously, supporting faster scanning time compared to discrete mode. The two slide rails support horizontal and vertical hlmovement, and are controlled by a



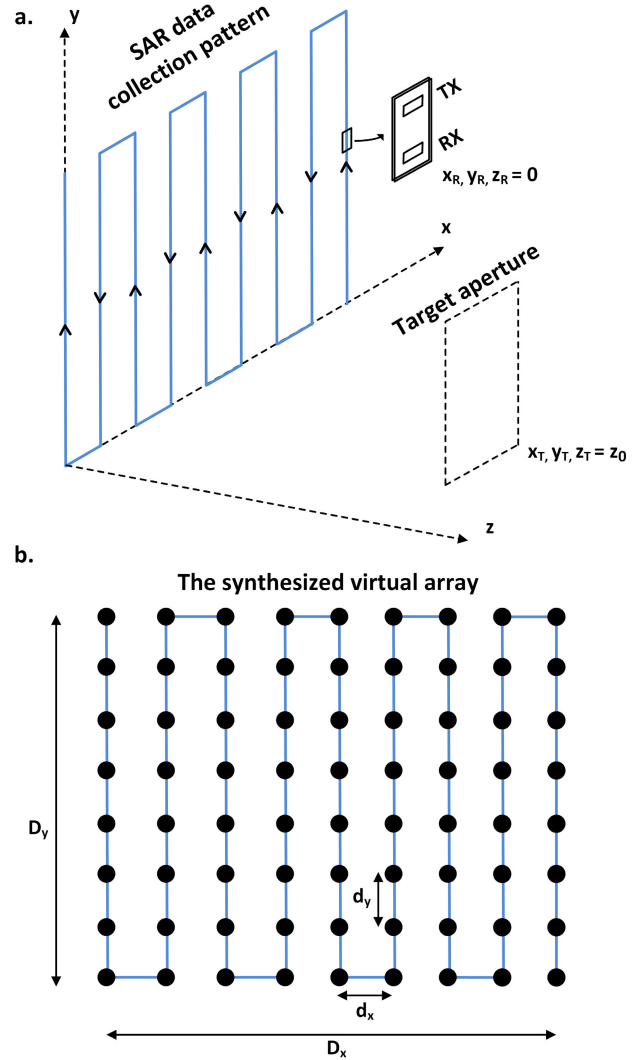


**FIGURE 3.** mmWave scanning & acquisition system. a) Block diagram of mmWave scanning system. b) mmWave scanning system containing TI AWR1843Boost and DCA1000 data capture.

motion controller, stepper motors, and their drivers. Finally, a laptop automatically controls the radar module, data capture module, and scanning platform through MATLAB software. Fig. 3b shows the real implementation of mmWave scanning system for near-field SAR imaging. This study utilized the single-input single-output (SISO) antenna mode with the TX1-RX2 and TX3-RX2 transceiver configurations, each employing different transmit and receive antennas. Therein, the TX and RX spacing in TX3-RX2 transceiver is greater than that of the TX1-RX2 transceiver.

### B. FMCW SIGNAL MODEL

To generate high-resolution images, a FMCW is utilized as the transmitted signal and a zig-zag scanning pattern is



**FIGURE 4.** The geometry of mmWave SAR system scanning. a) SAR collection pattern using zigzag pattern. b) The synthesized virtual array.

employed as shown in Fig. 4. A Cartesian coordinate system is established, denoting the horizontal, vertical, and range directions with the x-axis, y-axis, and z-axis, respectively. Assuming that a single target aperture is located at the position  $(x_T, y_T, z_0)$  with target reflectivity  $\rho(x_T, y_T)$ , and FMCW radar is positioned at  $(x_R, y_R, 0)$ , as shown in Fig. 4a. In this study, the data acquisition process is performed at all measurement points following the zigzag pattern and producing a synthesized virtual array, as illustrated in Fig. 4b. At each measurement point, the intermediate frequency (IF) signal is represented in the wavenumber domain, as follows:

$$m_{IF}(x, y, k) = \frac{\rho(x_T, y_T)}{R^2} \exp(j2\pi kR), \quad \frac{2\pi f_0}{c} \leq k \leq \frac{2\pi f_T}{c} \quad (1)$$

where  $R$  denotes the distance between the radar and the target aperture,  $f_0$  is the carrier frequency,  $f_T$  is the maximum swept

frequency of the FMCW signal, and  $k$  is the wavenumber, which corresponds to a specific frequency  $f$ .

The target reflectivity, denoted by  $\rho(x_T, y_T)$  is employed to characterize the 2D target at all measurement points. Consequently, all backscattered signals from the target aperture at a distance  $z_0$  can be expressed as follows:

$$m(x, y, k) = \iint \frac{\rho(x_T, y_T)}{R^2} \exp(j2kR) dx_T dy_T \quad (2)$$

Given that the radar is modeled as a single full-duplex antenna located equidistant between the transmitting (TX) and receiving (RX) antennas,  $R^{-2}$  can be approximated by  $(z_0 R)^{-1}$ . In the context of near-field imaging, which is the focus of this study, the error introduced by this approximation can be deemed negligible [42]. Furthermore, since the  $z_0$  remains constant during the scanning process, this value can be disregarded. Thus, the signal data utilized for image reconstruction is processed primarily through a combination of phase terms, along with the residual  $R^{-1}$  dependence. Then, Eq. (2) can be reformulated as:

$$m(x, y, k) = \iint \frac{\rho(x_T, y_T)}{R} \exp(j2kR) dx_T dy_T \quad (3)$$

To derive a high-resolution 2D SAR image, the scanning parameters must adhere to the Nyquist sampling criteria. Specifically, the scanning step sizes  $d_x$  and  $d_y$  should be smaller than center wavelength  $\lambda_c$ , which represents the worst-case scenario when the target is in close proximity to the radar sensor. The cross-range and range resolutions for near-field SAR imagery are described [10], [43], as follows:

$$\delta_x \approx \frac{\lambda_c z_0}{2D_x}, \delta_y \approx \frac{\lambda_c z_0}{2D_y}, \delta_z \approx \frac{c}{2B} \quad (4)$$

where  $D_x$  and  $D_y$  are scanning aperture sizes in the virtual array, with  $c$  being the speed of light and  $B$  being the bandwidth.

### C. RECONSTRUCTED IMAGING ALGORITHM

Various algorithms are available for reconstructing near-field SAR imagery from measurement points. In this study, the Range Migration Algorithm (RMA) was employed as the reconstruction algorithm, showcasing promise in real-time processing applications [44]. The aggregation of all measurement points produced a three-dimensional (3D) data cube essential for SAR image reconstruction. The dimensions of this data cube consist of the number of samples  $N_s$ , and the number of measurement points in the horizontal  $H_p$  and vertical  $V_p$  directions. Given  $m(a, b)$  as the signal received at a measurement point  $(x_a, y_b)$ , the formation of the data cube is expressed as:

$$\mathbf{M} = \begin{bmatrix} m(H_p, 1) & m(H_p, 2) & \cdots & m(H_p, V_p) \\ \vdots & \vdots & \ddots & \vdots \\ m(1, 2) & m(2, 2) & \cdots & m(2, V_p) \\ m(1, 1) & m(1, 2) & \cdots & m(1, V_p) \end{bmatrix} \quad (5)$$

The RMA transforms data in the cube using range-FFT, denoted as  $\Gamma(M)$ . Subsequently, range focusing to determine the target planar is performed at distance  $z_0$ , denoted as  $S = \Gamma(M, k_{z_0})$  with the dimension of  $H_p \times V_p$ . A matched filter can be expressed as:

$$MF = \exp\left(-j2k\sqrt{x^2 + y^2 + (z_0)^2}\right) \quad (6)$$

Finally, the near-field SAR imaging can be reconstructed using target planar distance  $z_0$  and a matched filter as:

$$\hat{M} = \text{IFT}_{2D}\{\text{FT}_{2D}(S \otimes MF)\} \quad (7)$$

where  $\otimes$  denotes the element-wise multiplication operator for matrices. In this equation, the Stolt interpolation was omitted because it could degrade imaging performance [45].

### D. NEAR-FIELD SAR IMAGING DESNOISING

To reduce noise in SAR images, various techniques are available, with the wavelet transform being widely acknowledged for its effectiveness [46]. Among these, DTCWT is recognized for its superior performance compared to other wavelet-based methods in far-field applications [47]. In this study, a denoising process based on DTCWT is implemented to enhance image quality and improve the contrast between defect and defect-free areas in near-field NDT applications.

The denoising process using the DTCWT for near-field SAR images is illustrated in Fig. 5. Initially, a noisy SAR image undergoes decomposition via forward DTCWT. This decomposition utilizes two filter trees, real part ( $h_0(n)$ ,  $h_1(n)$ ) and imaginary part ( $g_0(n)$ ,  $g_1(n)$ ), to extract wavelet and scale coefficients. Subsequently, a soft thresholding technique is applied to these coefficients to reduce noise effectively while preserving critical image features. After noise attenuation, the modified coefficients are reconstructed using an inverse DTCWT, employing the real ( $\overline{h_0(n)}$ ,  $\overline{h_1(n)}$ ) and imaginary ( $\overline{g_0(n)}$ ,  $\overline{g_1(n)}$ ) filter branches. The resultant denoised SAR image exhibits significantly reduced noise levels compared to the original noisy image. The thresholding step, utilizing SureShrink rules [48], is crucial for enhancing the signal-to-noise ratio, thereby improving the robustness of the mmWave imaging system against noise. Additionally, the Peak Signal-to-Noise Ratio (PSNR) [49] is employed to assess the SAR image quality between the original and denoised images, determining the efficacy of the denoising process.

### III. AUTOMATED DEFECT LOCALIZATION VIA IMAGE DECOMPOSITION

Although noise in near-field SAR images has been reduced, disturbances introduced by the scanning mechanism described in Sec. II-A still affect the generated images. These disturbances are typically non-smooth because the signal capture speed varies along different axes, leading to false defect identification during manual labeling. To automatically localize defects without manual labeling, Yan et al. have established the smooth-sparse decomposition (SSD) method [50]. This method separates sparse defects

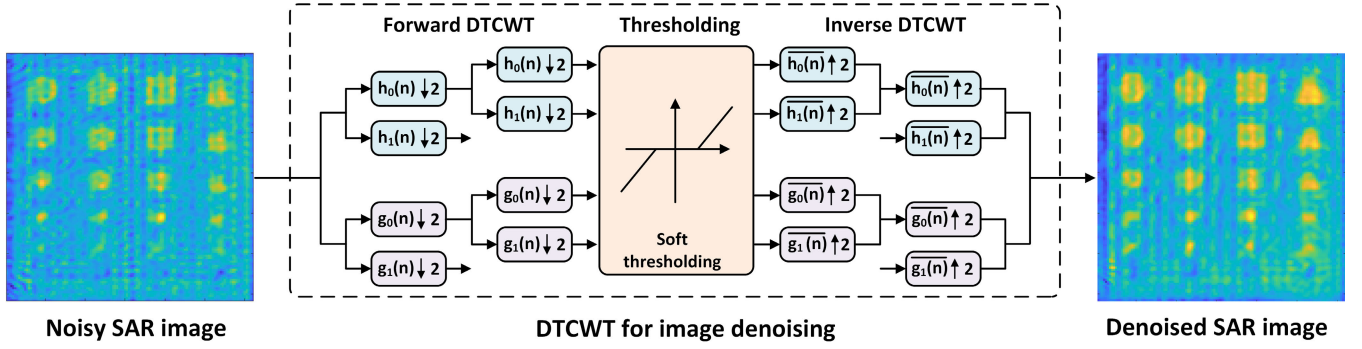


FIGURE 5. Dual-tree complex wavelet transform for near-field SAR image denoising.

from a smooth background using sharp and smooth functional bases. It assumes a smooth background pattern with sparse defects that exhibit distinct pixel intensity values. The defects can then be localized through the coefficients of the sharp functional basis. However, the intensity differences between the background disturbances and the defects become less distinguishable due to the large noise along the scanning path. Directly adopting the SSD method under these conditions can introduce false identifications in the detected defects.

To improve defect identification accuracy, we developed an automated algorithm that leverages scanning system settings to model and separate the complex disturbances in background patterns. The proposed algorithm aims to decompose an image  $\mathcal{Y}_{n_1 \times n_2}$  into three parts: background pattern containing the disturbances  $\mu$ , defect regions  $\eta$ , and random noises  $\epsilon$ . Each image  $\mathcal{Y}$  is represented as  $\mathcal{Y} = \mu + \eta + \epsilon$ , and is modeled via basis expansion as follows:

$$\mathcal{Y} = \mathcal{B}_\mu \theta_\mu + \mathcal{B}_\eta \theta_\eta + \epsilon, \quad (8)$$

where  $\mathcal{B}_\mu$  and  $\mathcal{B}_\eta$  are the 2D bases for the background and defect regions, respectively.

The difference between Eq. (8) and the formulation in [50] is that we modified the smooth background term to capture the pattern of disturbances. By modeling and separating the disturbance pattern, defects can be accurately identified despite the similar intensity values of the defects and the scanning path. The 2D bases  $\mathcal{B}_\mu$  and  $\mathcal{B}_\eta$  are determined by taking the tensor product of 1D bases, denoted as  $\mathcal{B}_\mu = B_{\mu,1} \otimes B_{\mu,2}$ . The coefficients are contained in  $\theta_\mu$  and  $\theta_\eta$ , respectively. The objective is to determine the optimal basis coefficient  $\theta_\eta$ , which indicates the locations of the anomalous regions in the image. For  $\theta_\eta$  to accurately capture the defects, precise modeling of both background disturbances and defects is required. The formulation for finding the optimal coefficients in Eq. (8) is as follows:

$$\begin{aligned} \arg \min_{\theta_\mu, \theta_\eta} \quad & \|\epsilon\|^2 + \lambda \theta_\mu^T R \theta_\mu + \gamma \|\theta_\eta\|_1 \\ \text{subject to} \quad & \mathcal{Y} = \mathcal{B}_\mu \theta_\mu + \mathcal{B}_\eta \theta_\eta + \epsilon, \end{aligned} \quad (9)$$

where  $\|\cdot\|$  and  $\|\cdot\|_1$  are  $L_2$  and  $L_1$  norm operators, and  $\lambda$  and  $\gamma$  are penalty parameters,  $R$  is defined as  $R = B_2^T B_2 \otimes$

$D_1^T D_1 + D_2^T D_2 \otimes B_1^T B_1 + \lambda D_2^T D_2 \otimes D_1^T D_1$  [50], [51]. The matrix  $D_i$  represents the first-order difference matrix in the  $i$ -th dimension.

To solve the optimization problem of Eq. 9, constructing basis functions to capture different patterns in the SAR images is a major challenge. Informed by the processes of scanning and image reconstruction, we introduce domain-informed basis functions for background patterns and defects. A greedy grid search is used to determine the hyperparameters to define the basis functions.

#### A. PROCESS-INFORMED BASIS FUNCTIONS TO MODEL SAR IMAGES

##### Bases for Modeling Background Patterns $\mathcal{B}_\mu$

The bases  $\mathcal{B}_\mu$  are chosen to represent the pattern of disturbances introduced by the scanning path, which are highly correlated to the scanning settings. Suppose the sensor is moving along a zig-zag path as presented in Fig. 4, at each given  $x$  location, the sensor moves continuously along the  $y$  axis. Correspondingly, the sensor moves discretely along the  $x$  axis at a given  $y$  location. This leads to continuous background patterns along the  $y$  axis and discrete patterns along the  $x$  axis. We choose  $N_{By}$  smooth bases, such as B-spline bases, and  $N_{Bx}$  sharp bases, such as Gaussian kernel bases, to capture the various patterns along the  $y$  axis and  $x$  axis, respectively. Since the number of discrete points along the  $x$  axis is highly correlated with the scanning parameters, we can determine  $N_{Bx}$  based on this information as:

$$N_{Bx} = \frac{D_x}{d_x} \cdot \frac{S_x}{I_x}, \quad (10)$$

where  $S_x$  is the length of the part and  $I_x$  is the image size along the  $x$  axis. The  $N_{By}$  is determined by an optimal grid search to accommodate different smooth background patterns along the  $y$  axis.

##### Bases for Modeling Anomaly $\mathcal{B}_\eta$

Since the target is to identify the sizes and locations of the defects, we need bases that clearly delineate their boundaries and precisely identify their locations. Using B-spline bases or thin-plate regression spline bases may not clearly determine their sizes since their widths are determined by the specified

number of bases. Therefore, we choose Gaussian kernel bases to model the functions in  $\mathcal{B}_\eta$ . A Gaussian kernel basis in one dimension of  $\mathcal{B}_\eta$  can be defined as:

$$B_{\eta,i}(v) = \exp\left(-\frac{(v - v_{it})^2}{d_{it}}\right), \quad (11)$$

where  $v_{it}$  is the location of the  $t = 1, \dots, N_G$  knot. Knots are uniformly placed within the range of  $v$ . The length-scale parameter  $d_{it}$  controls the kernel width and effective range to model the signal data. A higher value of  $d_{it}$  induces a flatter Gaussian kernel, which is less sensitive to signal noise and peaks. Suppose  $N_{Ax}$  and  $N_{Ay}$  are the number of Gaussian kernel bases used to identify the defects along the  $x$  and  $y$  axes, respectively.

### B. HYPERPARAMETER OPTIMIZATION VIA GRID SEARCH

Since the sizes of the defects are typically unknown, the choice of the width and number of functional bases must adapt to this variability. We adopt a grid search scheme to determine the optimal hyperparameters for fitting background and defect patterns in images. It systematically explores combinations of hyperparameters, including the number of knots and widths for both background  $\mathcal{B}_\mu$  and defect  $\mathcal{B}_\eta$  fitting functions. Specifically, the grid search iterates over a predefined range of values for the number of bases along the  $x$ -axis for modeling the disturbance ( $N_{Bx}$ ), the number of bases for modeling the defects ( $N_{Ax}$ ,  $N_{Ay}$ ), and their widths ( $d_{xt}$ ,  $d_{yt}$ ). For each combination, the code constructs basis functions for both background and defect patterns using B-splines and exponential basis functions. The performance of each hyperparameter set is evaluated based on detection accuracy, calculated from the decomposition results. The best-performing set of hyperparameters is identified by comparing the accuracy across all iterations, thereby automating the hyperparameter tuning process for optimal image pattern fitting.

The algorithm for solving Eq. 9 using a specified functional basis in one iteration of the grid search is summarized as in Algorithm 1. More details can be found in [50].

## IV. EXPERIMENT SETUP AND RESULTS: INTERNAL DEFECTS DETECTED ON 3D-PRINTED OBJECTS

This section demonstrates the effectiveness of the established mmWave SAR imaging system for detecting internal defects. The experimental setup is shown in Fig. 6, which includes both scheme view and real view. In this paper, we used PLA 3D-printed samples as the material under test (MUT) [28]. Each experiment includes MUTs with both defect-free and defective regions, as depicted in Fig. 6a. The defective regions are represented as voids within the 3D-printed structure. During inspection, the backside of the defective regions is covered with a 3D-printed cover of the same sample size to emulate internal defects. Fig. 6.b shows the actual experimental setup with the detailed experimental parameters listed in Table 3. The virtual array synthesis spans from -100 mm to 100 mm on both vertical and horizontal scanning

### Algorithm 1 Adaptive Decomposition of SAR Images

**Require:** Input SAR image  $Y$ , Basis type  $\mathcal{B}_{\eta,1}, \mathcal{B}_{\eta,2}$  to capture the defects and  $\mathcal{B}_{\mu,1}, \mathcal{B}_{\mu,2}$

**Ensure:**  $\theta_\eta$

- 1: Let  $Z_i Z_i^T = \mathcal{B}_{\mu,i}^T \mathcal{B}_{\mu,i}$ ,  
 $U_i \text{diag}(s_i) U_i^T = Z_i^{-1} D_i^T D_i (Z_i^{-1})^T$ ,  
 $V_i = B_i (Z_i^{-1})^T U_i, i = 1, 2$ ,  
 $L = 2 \|\mathcal{B}_{\eta,1}\|_2^2 \|\mathcal{B}_{\eta,2}\|_2^2, \theta_\eta^{(0)} = 0, X^{(0)} = 0, t_1 = 1,$   
 $k = 1$
- 2: **while**  $\|\theta_\eta^{(k-1)} - \theta_\eta^{(k)}\| > \varepsilon$  **do**
- 3:   Select  $\lambda^{(k)} = \arg\min_\lambda GCV(\lambda)$
- 4:    $H_i^{(k)} = H_i(\lambda^{(k)}) = V_i^T (I + \lambda^{(k)} \text{diag}(s_i))^{-1} V_i, i = 1, 2$
- 5:    $A^{(k)} = \mathcal{B}_{\eta,1} X^{(k-1)} \mathcal{B}_{\eta,2}^T$
- 6:    $M^{(k)} = H_1^{(k)} (Y - A^{(k-1)}) H_2^{(k)} \mathbb{I}_{\{L_l, L_u\}}$
- 7:    $\theta_e^{(k)} := X^{(k-1)} + \frac{2}{L} \mathcal{B}_{\eta,1}^T (Y - M^{(k)} - A^{(k)}) \mathcal{B}_{\eta,2}$
- 8:   Select  $\gamma^{(k)}$  using Otsu's algorithm:  $\gamma^{(k)} = \text{otsu}(\theta_e^{(k)}) \times L$
- 9:    $\theta_\eta^{(k)} = S_\gamma^{(k)}(\theta_e^{(k)})$
- 10:   Update  $t_k = \frac{1 + \sqrt{1 + 4t_{k-1}^2}}{2}$
- 11:   Update  $X^{(k)} = \theta_\eta^{(k-1)} + \frac{t_{k-1} - 1}{t_k} (\theta_\eta^{(k-1)} - \theta_\eta^{(k-2)})$
- 12:   Update  $k = k + 1$
- 13: **end while**

dimensions. To satisfy the sampling criterion, the scanning step was set to be less than the wavelength  $\lambda_c$ , with values of 0.8 mm selected for both the horizontal and vertical directions. The distance between the scanning aperture of radar sensor and MUTs ( $z_0$ ) was set at three values such as 30 mm, 50 mm, and 100 mm.

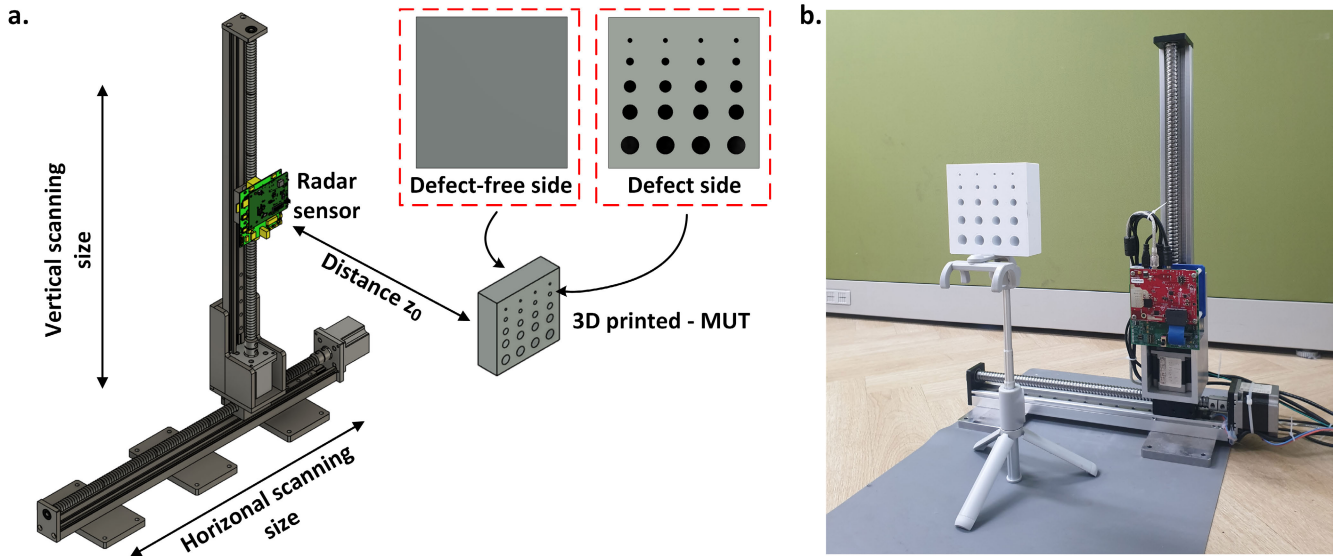
To assess the efficacy of the proposed framework, we conducted three sets of MUTs:

- **Shape and size variation:** Tested using multi-shape & multi-size samples.
- **Material permittivity and size variation:** Examined with multi-infill & multi-size samples.
- **Depth and size variation:** Evaluated using multi-depth & multi-size samples.

The dimensions and specifications of these MUTs can be found in Table 4.

The performance of the detection systems was evaluated using quantitative metrics, including Accuracy, False Positive Rate (FPR), and False Negative Rate (FNR). In particular, accuracy is defined as the ratio of correctly classified pixels to the total number of pixels, and can be expressed as follows: Accuracy =  $\frac{TP+TN}{TP+TN+FP+FN}$ , where  $TP$  (True Positives) represents the number of pixels correctly classified as defects,  $TN$  (True Negatives) represents the number of pixels correctly classified as non-defects,  $FP$  (False Positives) represents the number of pixels incorrectly classified as defects, and  $FN$  (False Negatives) represents the number





**FIGURE 6.** Non-destructive experiment setup for automated defect identification and localization. a) Scheme view of experiment setup, b) Real view of experiment setup.

of pixels incorrectly classified as non-defects. The FPR measures the proportion of non-defect pixels that are incorrectly classified as defects and is given by:  $FPR = \frac{FP}{FP+TN}$ . Similarly, the FNR measures the proportion of defect pixels that are incorrectly classified as non-defects and is calculated as follows:  $FNR = \frac{FN}{FN+TP}$ .

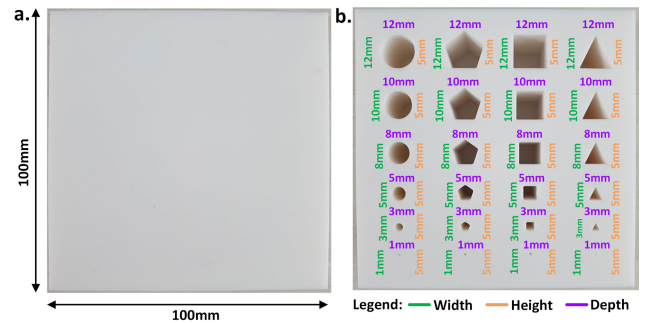
In the following sections, the synthesized defects were detected using the established mmWave SAR image system. These images were produced by transceivers TX1-RX2, TX3-RX2 as mentioned in Sec. II-A. And the SAR images are generated at a focus distance of  $z_0 + 1$  mm. This distance is chosen to demonstrate that the proposed algorithm can accurately detect defect locations without knowing their exact depth. We detected the defects from the generated SAR images using the proposed process-informed smooth-sparse decomposition (PISSD). The proposed method is compared with the regular smooth sparse decomposition (SSD) developed by Yan et al. [50]. For SSD method, we tested two sets of parameter settings: the first setting, denoted as SSD1, uses the same type and number of bases for anomalies,  $B_\eta$ , as found by grid searching in PISSD. The second setting, denoted as SSD2, uses the same configuration as in [50]. Additionally, we compare PISSD with a global thresholding method, specifically Otsu's method, denoted as Global.

#### A. EXPERIMENT 1: SHAPE AND SIZE VARIATION TEST

Real-world materials often contain defects of diverse geometries and dimensions. In this study, a 3D-printed multi-shape & multi-size MUT was designed with defect shapes including circles, pentagons, squares, and triangles as shown in Fig. 7. The MUT has dimensions of 100 mm x 100 mm and a thickness of 30 mm. The sample contains four types of defect

**TABLE 3.** The parameters of mmWave SAR scanning system.

| Parameter                                  | Value       |
|--------------------------------------------|-------------|
| Center Frequency (GHz)                     | 79          |
| Bandwidth (GHz)                            | 4           |
| Platform speed (mm/s)                      | 20          |
| Horizontal step size (mm)                  | 0.8         |
| Vertical step size (mm)                    | 0.8         |
| Synthetic aperture size (mm <sup>2</sup> ) | 200 x 200   |
| Distance between MUT and radar (mm)        | 30, 50, 100 |

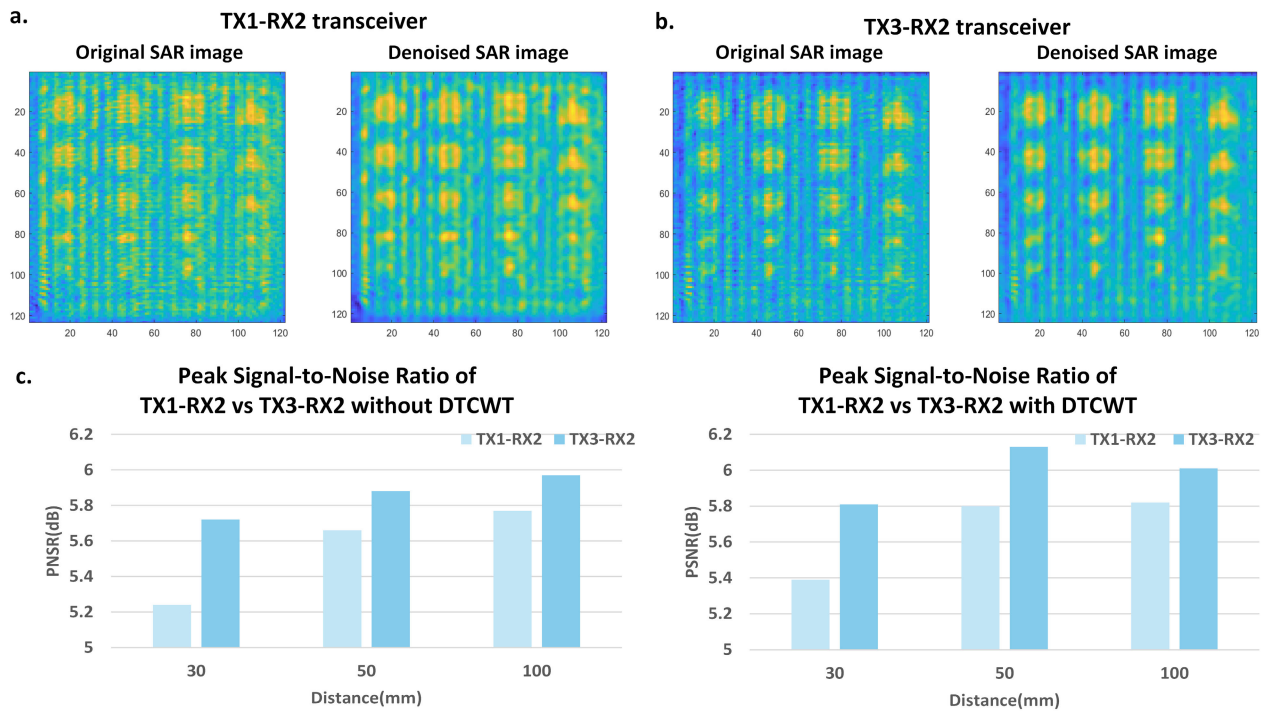


**FIGURE 7.** 3D-printed MUT sample: multi-shape & multi-size. a) The defect-free side - front side, b) The defect side - back side with the size and depth of defects.

shapes: circle, pentagon, square, and triangle. The defect-free side is illustrated in front side, as shown in Fig. 7a. These defects have a fixed depth value of 5 mm and various defect sizes from 12 mm x 12 mm to 1 mm x 1 mm, as depicted in Fig. 7b. This experiment assessed the performance of proposed automated detection framework across a multi-shape & multi-size object using a near-field SAR mmWave imaging system.

**TABLE 4.** Parameters of Testing Sample.

| Testing sample            | Parameter       |                  |               |              |                                          |                   |
|---------------------------|-----------------|------------------|---------------|--------------|------------------------------------------|-------------------|
|                           | Sample size     | Sample thickness | Defect size   | Defect depth | Defect shape                             | Infill percentage |
| Multi-shape & multi-size  | 100 mm x 100 mm | 30 mm            | 12 mm x 12 mm | 5 mm         | Circle<br>Pentagon<br>Square<br>Triangle | 50%               |
|                           |                 |                  | 10 mm x 10 mm |              |                                          |                   |
|                           |                 |                  | 8 mm x 8 mm   |              |                                          |                   |
|                           |                 |                  | 5 mm x 5 mm   |              |                                          |                   |
|                           |                 |                  | 3 mm x 3 mm   |              |                                          |                   |
|                           |                 |                  | 1 mm x 1 mm   |              |                                          |                   |
| Multi-infill & multi-size | 100 mm x 100 mm | 30 mm            | 12 mm x 12 mm | 5 mm         | Circle                                   | 65%               |
|                           |                 |                  | 10 mm x 10 mm |              |                                          | 50%               |
|                           |                 |                  | 8 mm x 8 mm   |              |                                          | 35%               |
|                           |                 |                  | 5 mm x 5 mm   |              |                                          | 20%               |
|                           |                 |                  | 3 mm x 3 mm   |              |                                          |                   |
| Multi-depth & multi-size  | 100 mm x 100 mm | 30 mm            | 12 mm x 12 mm | 10 mm        | Circle                                   | 50%               |
|                           |                 |                  | 10 mm x 10 mm | 8 mm         |                                          |                   |
|                           |                 |                  | 8 mm x 8 mm   | 5 mm         |                                          |                   |
|                           |                 |                  | 5 mm x 5 mm   | 3 mm         |                                          |                   |
|                           |                 |                  | 3 mm x 3 mm   |              |                                          |                   |
|                           |                 |                  |               |              |                                          |                   |



**FIGURE 8.** Denosing performance of DTCWT at across various testing distance at 30, 50, 100 mm at TX1-RX2 and TX3-RX2 transceiver. a) Original and denoised SAR images at testing distance 30 mm for TX1-RX2. b) Original and denoised SAR images at testing distance 30 mm for TX3-RX2. c) PSNR comparison without (left) and with (right) DTCWT denoising across different distances.

We also tested and compared the performance of applying denoising algorithm (DTCWT) for detecting defects with varied sizes and shapes. The denoising performance of DTCWT across various testing distance at 30, 50, and

100 mm at TX1-RX2 and TX3-RX2 transceiver is presented in Fig. 8. In general, DTCWT effectively removes noise from near-field SAR images, thereby enhancing image quality. In both Fig. 8a and Fig. 8b, the denoised SAR images exhibit

more pronounced contrast between defect and non-defect areas compared to the original SAR images. This indicates significant noise reduction and enhanced image clarity post-denoising, despite the persistence of vertical scanning patterns. In Fig. 8c, the right graph consistently shows higher PSNR values compared to the left graph across different distances and transceiver configurations, indicating improved image quality with DTCWT denoising. The TX1-RX2 transceiver has a lower PSNR than the TX3-RX2 transceiver. This difference in PSNR between these two transceivers may be attributed to the shorter distance between RX and TX channels in the TX1-RX2 transceiver as shown in Fig. 3, leading to higher interference coupling noise [32], [33]. In brief, these results emphasize the utility of DTCWT in providing clearer near-field SAR images with higher PSNR, thereby supporting the defect detection accuracy, despite the enduring presence of vertical scanning patterns. Moreover, in configuring mmWave system parameters, it's important to consider transceiver distance to minimize unwanted noise.

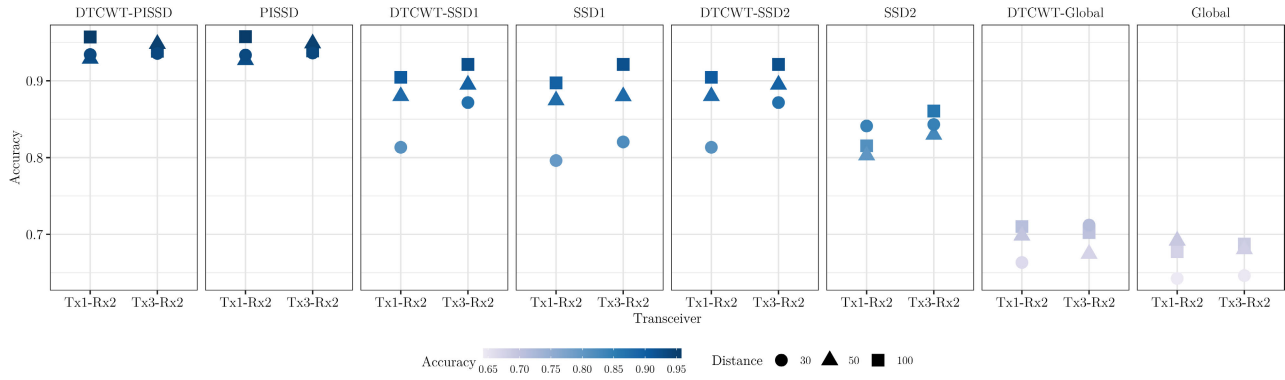
To evaluate the impact of DTCWT on image decomposition methods, Fig. 9 compares the accuracy of these methods in detecting internal defects. The image decomposition methods, including PISSD, SSD2, SSD1, and Global, were evaluated both with and without the application of DTCWT. The DTCWT-enhanced methods include DTCWT-PISSD, DTCWT-SSD1, DTCWT-SSD2, and DTCWT-Global. These evaluations were performed at distances of 30, 50, and 100 mm using TX1-RX2 and TX3-RX2 transceivers. The results show that most algorithms utilizing the DTCWT exhibit higher accuracy across different transceiver configurations and testing distances compared to those without DTCWT. In terms of transceiver configuration, the TX3-RX2 setup consistently achieves higher accuracy than the TX1-RX2 setup, particularly in methods such as DTCWT-SSD1, SSD1, DTCWT-SSD2, and SSD2. This improvement is attributed to the higher PSNR values observed in previous experiments. The analysis further demonstrates that the PISSD and DTCWT-PISSD algorithms consistently achieve over 0.9 accuracy in detecting internal defects across all tested distances, outperforming other methods. Notably, the PISSD algorithm performs comparably to DTCWT-PISSD in all transceiver configurations. In contrast, other methods require the application of the DTCWT filter to achieve higher accuracy. These findings suggest that proposed automated method (PISSD) is also capable of eliminating noise in near-field SAR images. In summary, DTCWT enhances the performance of image decomposition in internal defect detection across different transceiver configurations and various testing distances.

The effectiveness of the proposed image decomposition method (PISSD) was further evaluated using a multi-shape & multi-size test, as illustrated in Fig. 10. The MUT was positioned at a 100 mm testing distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT. The results show that in detected defects image, the internal defects can be clearly

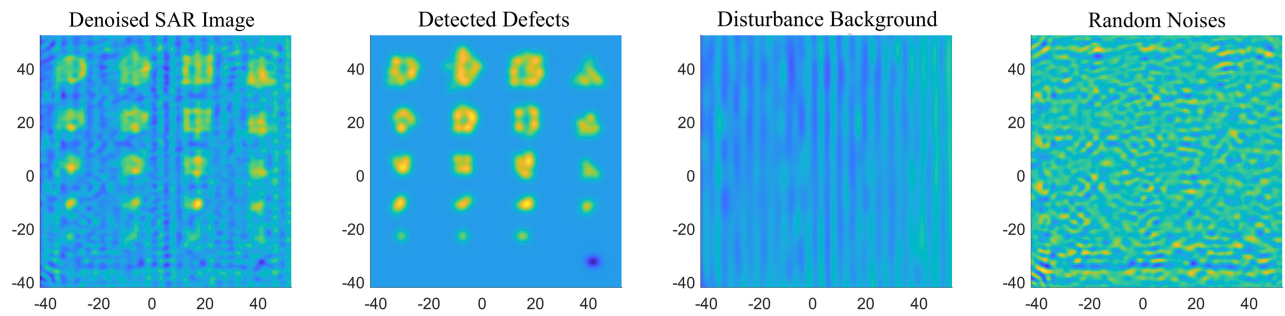
identified as circle, pentagon, square, and triangle shapes for most defect sizes. Notably, defects of  $3 \text{ mm} \times 3 \text{ mm}$  are successfully detected, although they are challenging to recognize in the denoised SAR image due to the influence of disturbance patterns and noise. However, at a defect size of  $1 \text{ mm} \times 1 \text{ mm}$ , only one triangle defect can be recognized. This is because, as the defect size decreases, the reflected area also decreases, leading to less reflected signal from the defects. The vertical scanning patterns are distinctly captured in the disturbance background image by leveraging scanning system settings, while the random noise image also reveals residual noise in the near-field SAR image. Overall, these findings suggest that the proposed image decomposition (PISSD) effectively models and separates boundaries of defect patterns in various shapes, sizes as well as the disturbances patterns of vertical scanning and noise.

The comparative performance of the PISSD, SSD1, SSD2, and Global methods is presented in Fig. 11, focusing on the detection of defects in a multi-shape & multi-size test. The MUT was positioned at a 100 mm testing distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT. The PISSD method clearly identifies defects of various shapes and sizes with fewer falsely detected defects. Because this method effectively models and separates defect patterns, reducing the false defect detection rate. In the SSD methods, the SSD2 method presents better defect identification than SSD1 method. But both still contain unwanted noise such as white dots. The Global method performs poorly, as it fails to distinguish between disturbance patterns and defect patterns. In summary, this analysis demonstrates that PISSD method outperforms existing methods in classifying internal defects of multiple shapes and sizes out of disturbances and random noises, significantly differentiating between defective and non-defective regions.

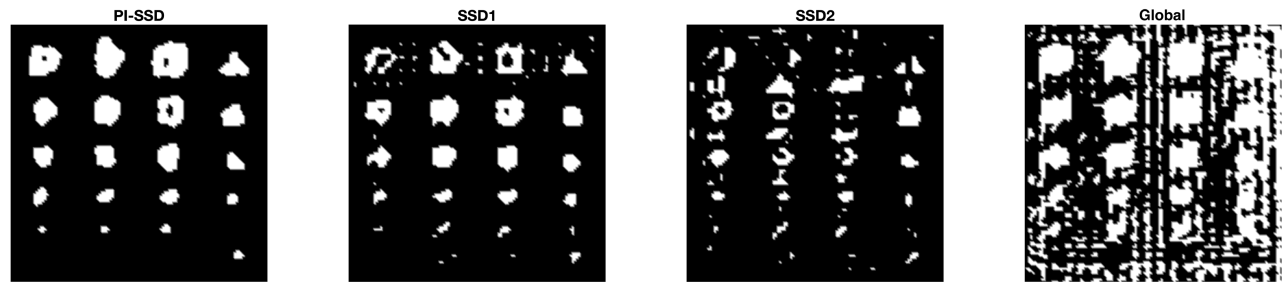
Finally, a comprehensive analysis of image decomposition methods reveals the superior performance of the proposed framework, particularly the PISSD method, across various defect shapes, sizes, and testing distances. The evaluation metrics demonstrate that the proposed method consistently outperforms other approaches in almost all categories, as shown in Fig. 12. For example, at a distance of 50 mm, the PISSD algorithm achieves the highest accuracy, with values ranging from 0.925 to 0.966. While the DTCWT-PISSD algorithm achieves the second-highest accuracy, with values ranging from 0.935 to 0.957, showing a smaller value gap. At other distances, although there is a slight decrease in accuracy, both PISSD and DTCWT-PISSD still maintain high performance, demonstrating their robustness. Regarding shape defects, the best FNR is achieved by DTCWT-PISSD (0.176) for the pentagon shape, and the best FPR is achieved by PISSD (0.013) for the triangle shape. The FPR remains consistently low across all distances and shapes, with the DTCWT-PISSD method ranging from 0.014 to 0.050 and the PISSD method ranging from 0.013 to 0.038, indicating high reliability in defect detection without false alarms.



**FIGURE 9.** Comparison of the accuracy of using different methods and two transceiver configurations at across various testing distance at 30, 50, 100 mm.



**FIGURE 10.** Proposed image decomposition for multi-shape & multi-size test. The MUT was positioned at a 100 mm distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT.



**FIGURE 11.** Detected defects using different methods for multi-shape & multi-size test. The MUT was positioned at a 100 mm distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT.

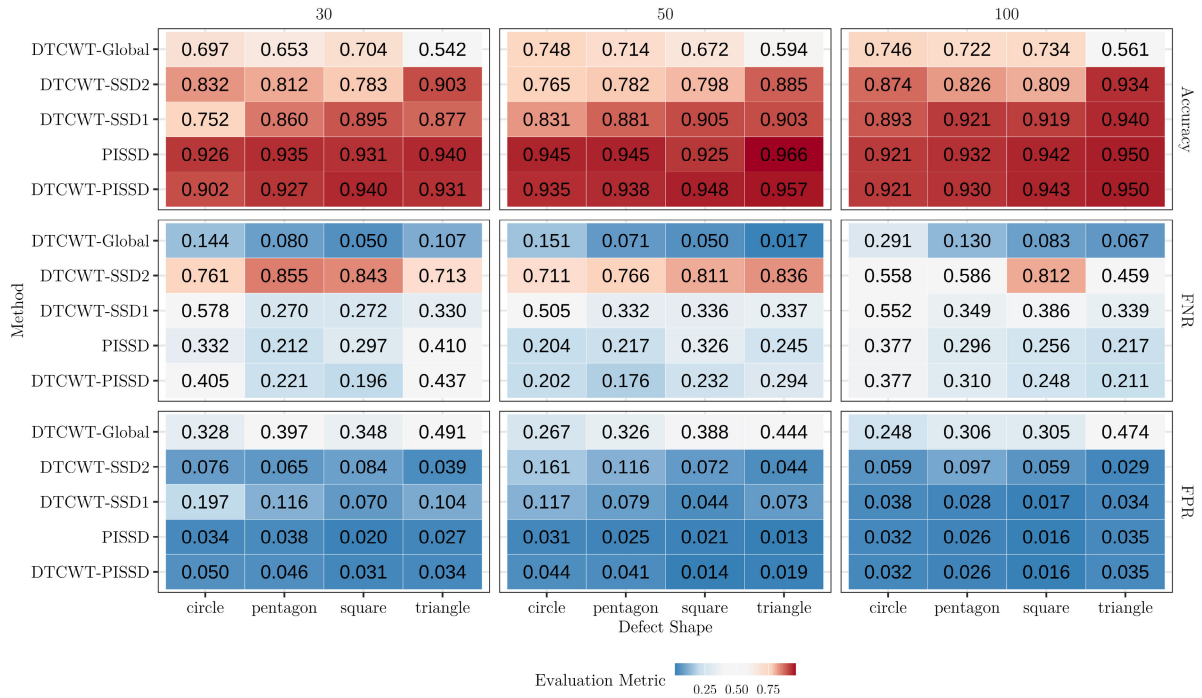
The DTCWT-Global algorithm shows the best average FNR of 0.103 in all cases. However, DTCWT-Global algorithm has the lowest average accuracy (0.674) and the highest average FPR (0.360), indicating its limited robustness. In summary, the proposed framework, particularly PISSD method, demonstrates robustness and effectiveness across different defect shapes, sizes, and testing parameters in near-field SAR mmWave NDT systems.

## B. EXPERIMENT 2: MATERIAL PERMITTIVITY AND SIZE VARIATION TEST

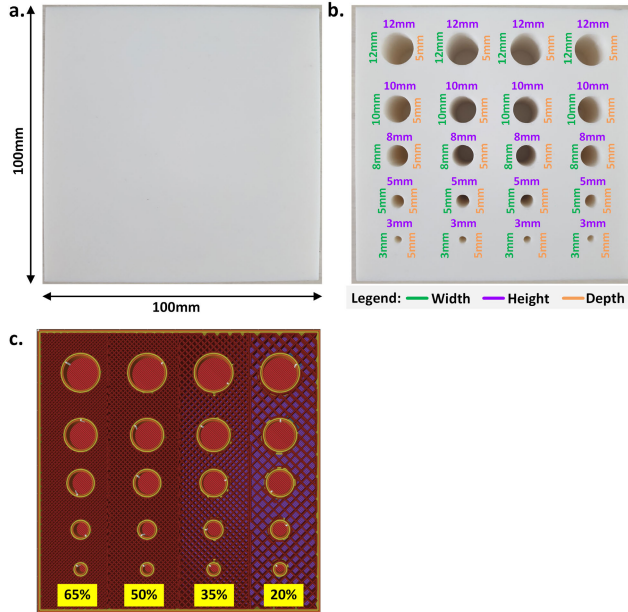
Material permittivity plays a crucial role in mmWave NDT imaging systems as it governs the interaction of electromagnetic wave in material [52], [53]. In PLA 3D

printing material, higher material permittivity is associated with higher infill percentages, while lower infill percentages result in reduced permittivity [54]. To assess the impact of material permittivity on defect detection, this experiment used 3D-printed multi-infill & multi-size MUTs with varying infill percentages. Fig. 13 presents the design profile of the MUT, including front and back sides. The MUT incorporates four distinct infill percentages: 65%, 50%, 35%, and 20%. These percentages correspond to different material permittivity levels and allow for systematic evaluation of their influence on defect detecting system. Similar to the multi-shape & multi-size MUTs, the multi-infill & multi-size MUTs have consistent dimensions: 100 × 100 mm size, 30 mm thickness, and 5 mm defect depth. The defect sizes





**FIGURE 12.** Comprehensive analysis with evaluation metric: Accuracy, FNR, FPR of proposed method DTCWT-PISSD, PISSD and other methods across different shapes (circle, pentagon, square, and triangular) in the multi-shape & multi-size tests. The distance between radar sensor and MUTs ( $z_0$ ) is 30 mm, 50 mm and 100 mm, respectively.

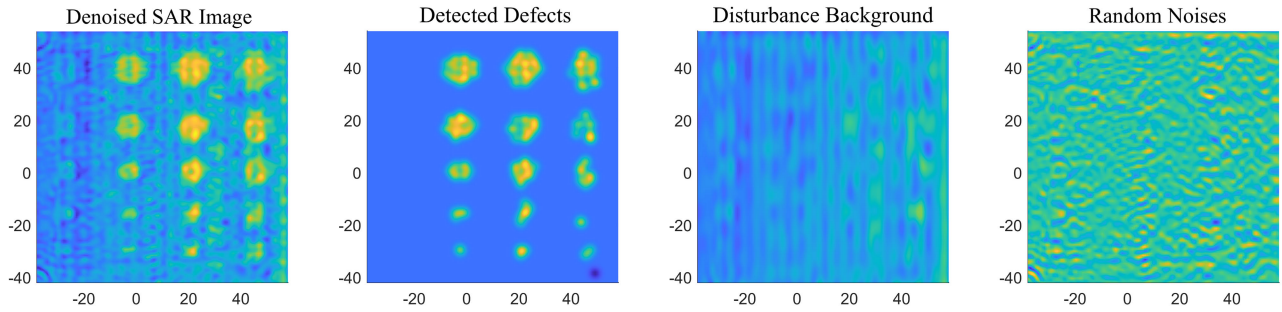


**FIGURE 13.** 3D-printed MUT sample: multi-infill & multi-size. a) The defect-free side - front side, b) The defect side - back side with the size and depth of defects, c) The local region with various infill percentage: 65%, 50%, 35% and 20%.

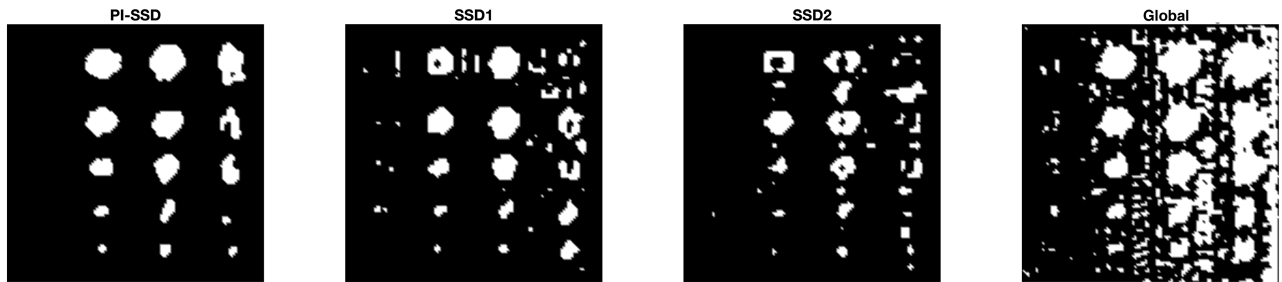
range from 12 mm  $\times$  12 mm to 3 mm  $\times$  3 mm, all exhibiting with a circular shape, as summarized in Table 4. This experiment evaluated the efficacy of the proposed automated detection framework under varying material permittivity levels and defect sizes.

The results of the proposed image decomposition method (PISSD) in multi-infill & multi-size test are illustrated in Fig. 14. The MUT was positioned at a 100 mm testing distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT. In the denoised SAR image, the differences in background noise and amplitude of defects can be observed in four regions due to variations of infill percentage values or material permittivity levels. Specifically, the amplitude of defects are highest in 35% infill region, and lowest value in 65% infill region. In the highest material permittivity region (65% infill), the circle defects show no visualized in detected defects image. Besides the reasons relating to disturbance, noise and size of defects, it suggests that high permittivity materials also obscure defects, reducing penetration capability of these materials in mmWave NDT systems. However, in remaining regions of detected defects image, circle defects ranging from 12 mm  $\times$  12 mm to 3 mm  $\times$  3 mm are still successfully identified. In the disturbance background image, the vertical scanning pattern is captured and its density unevenly distributed, showing higher amplitude in areas with lower material permittivity. These findings imply that the proposed image decomposition method (PISSD) has decomposition ability to separate anomaly regions, disturbance patterns, and noises in materials having variations of permittivity, enhancing the identification of defect boundaries.

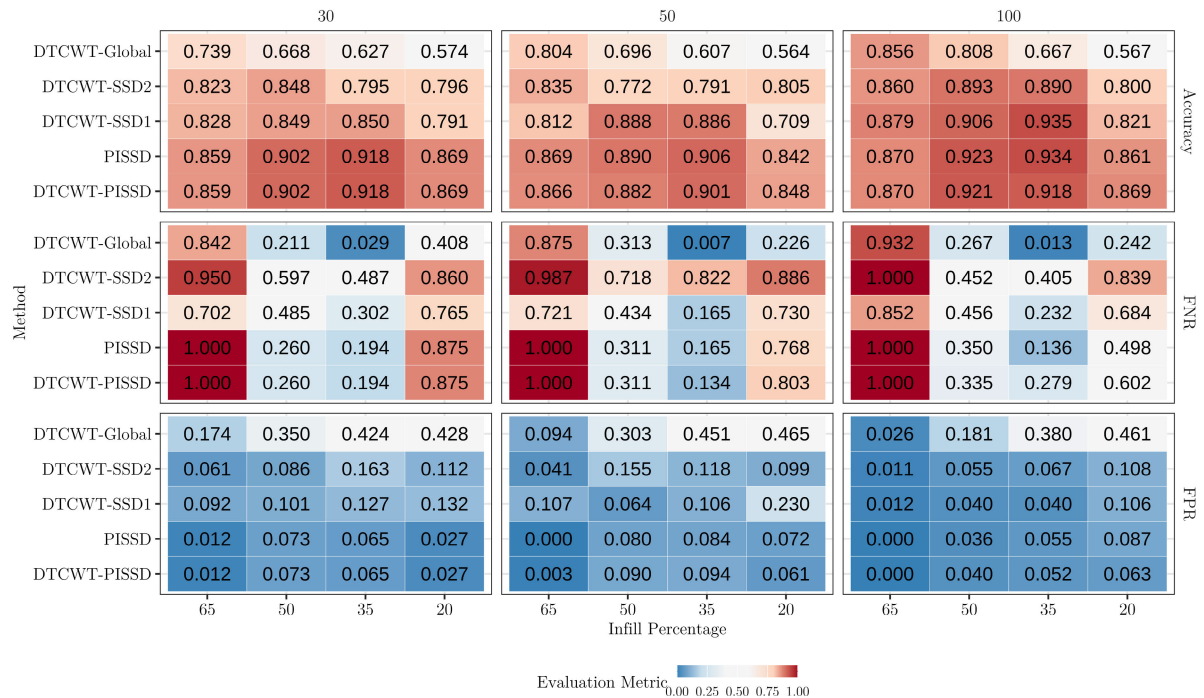
To demonstrate the superiority of proposed image decomposition method (PISSD) for defect detection in a multi-infill



**FIGURE 14.** Proposed image decomposition for multi-infill & multi-size test. The MUT was positioned at a 100 mm distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT.



**FIGURE 15.** Detected defects using different methods for multi-infill & multi-size test on generated raw images. The MUT was positioned at a 100 mm distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT.



**FIGURE 16.** Comprehensive analysis with evaluation metric: Accuracy, FNR, FPR of proposed method DTCWT-PISSD, PISSD and other methods across different local regions with 65%, 50%, 35%, 20% infills in the multi-infill & multi-size tests. The distance between radar sensor and MUTs ( $z_0$ ) is 30 mm, 50 mm and 100 mm, respectively.

& multi-size test, PISSD was compared with SSD1, SSD2, and Global methods, as depicted in Fig. 15. The MUT was

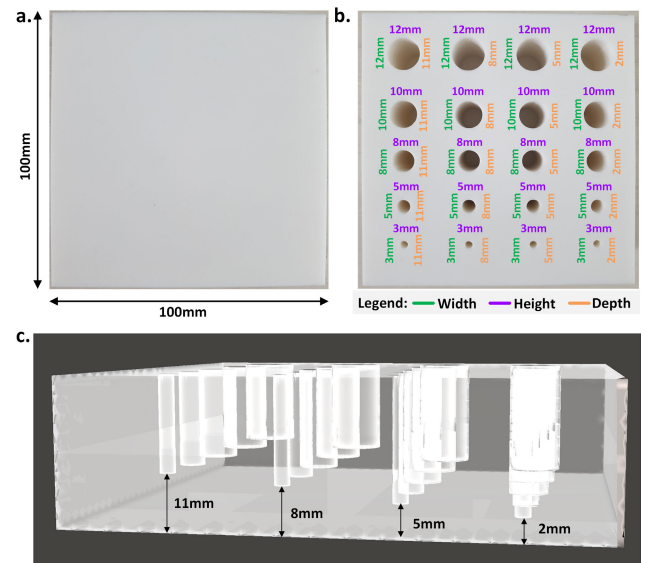
positioned at a 100 mm testing distance from the radar sensor configured with TX3-RX2 transceiver. The image was

preprocessed using DTCWT. Generally, in regions with the highest material permittivity (65% infill region), none of the methods detect the defects. In remaining regions, SSD1 and SSD2 perform better than the Global method, but undesired defects are still present. Additionally, the defects identified by SSD1 and SSD2 methods in the 20% infill region are not as clearly shown compared to those in the 50% and 35% infill regions. This is due to the low permittivity in the 20% infill region, resulting in reduced mmWave signal reflection [53]. In contrast, the PISSD method outperforms the others by clearly identifying defects across most infill percentage regions, even in lowest permittivity region. Therefore, the proposed image decomposition method (PISSD) demonstrates superior capability in classifying defects in materials with varying permittivity characteristics, including low permittivity materials, compared to other methods.

For comprehensive analysis, image decomposition methods were evaluated using metrics across infill percentages and sizes of internal defects at three testing distances, as presented in Fig. 16. The results indicate that both the proposed defect detection algorithms consistently achieve high accuracy and low rates of FNR and FPR, outperforming other methods in all local regions and testing distances. The accuracy of both PISSD and DTCWT-PISSD algorithms is similar across all local regions, with values ranging from 0.842 to 0.934. For all methods, low accuracy and high FNR are observed in local regions with infill percentages of 65% and 20% due to the low amplitude of reflected signals in these regions. However, in 20% infill region, PISSD still shows the lowest FNR(0.498) compared to other methods at a 100 mm testing distance, expect DTCWT-Global. In the remaining local regions, the proposed algorithms achieve a good balance between FNR and FPR, with these values being small and having minimal differences, such as FNR: 0.260 - FPR: 0.073 in 50% infill region and FNR: 0.194 - FPR: 0.065 in 35% infill region at a 30 mm testing distance. The FPR is notably low across all methods and infill percentages, with the DTCWT-PISSD and PISSD algorithms achieving minimal false alarms ranging from 0.000 to 0.094. These results confirm that the proposed defect detection algorithms are especially effective and reliable across local regions with various infill percentages and sizes, indicating that they are robust in materials with varying permittivity characteristics in near-field SAR mmWave NDT systems.

### C. EXPERIMENT 3: DEPTH AND SIZE VARIATION TEST

The depth of internal defects significantly impacts the detectability of mmWave imaging systems, as deeper defects result in weaker reflected signals and greater challenges for accurate detection. To evaluate the ability of the proposed automated detection framework, a 3D-printed multi-depth & multi-size MUT was designed with varying defect depths and sizes. Fig. 17 shows MUT features consistent dimensions, with a size of 100 mm  $\times$  100 mm and a thickness of 30 mm, similar to the two previously tested MUTs. However, this MUT incorporates circular defects with depths ranging



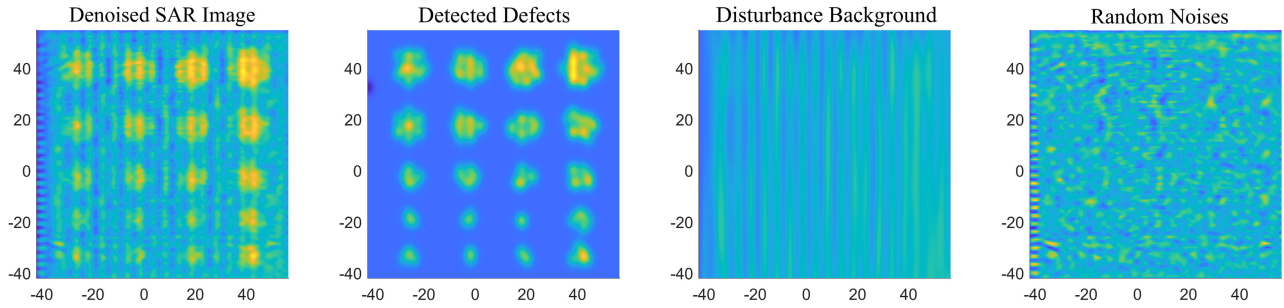
**FIGURE 17.** 3D-printed MUT sample: multi-depth & multi-size. a) The defect-free side – front side, b) The defect side – back side with the size and depth of defects, c) The inside view of MUT with various depths.

from 10 mm to 3 mm, as detailed in Table 4. By varying defect depths, the experiment assesses the effectiveness of the proposed framework in detecting internal defects under different depth conditions.

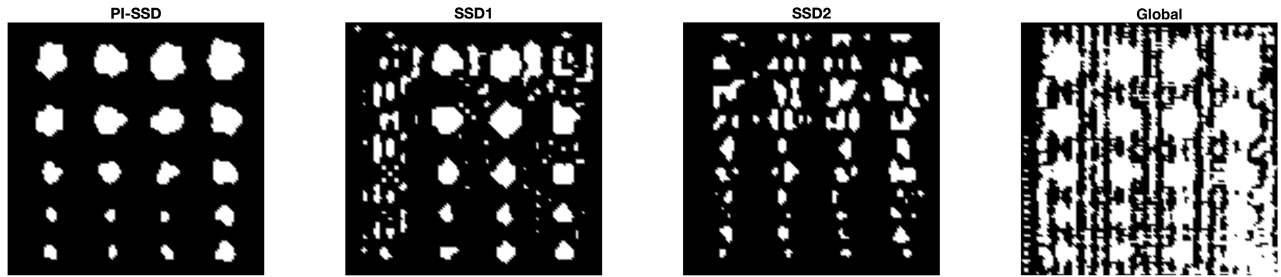
The performance of proposed image decomposition method (PISSD) is presented for its effectiveness in handling multi-depth & multi-size scenarios, as shown in Fig. 18. The MUT was positioned at a 30 mm testing distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT. In a denoised SAR image, smaller and deeper defects show smaller amplitude. For example, at a defect depth of 11 mm, the amplitude of defects is lower with smaller defect sizes, making them difficult to identify and localize. This problem is caused by not only noisy disturbance, size of defects but also by depth of defects, making internal defect detection challenging. However, in defected-defects image, defects of all depths and sizes are clearly visualized. Vertical scanning patterns and noises are captured in disturbance background and random noise images, respectively. In summary, these results indicate that proposed method (PISSD) has capability to model and separate defects in multi-depth & multi-size cases.

The comparison among PISSD, SSD1, SSD2, and Global methods is shown for detecting defects in a multi-depth & multi-size test in Fig. 19. The MUT was positioned at a 30 mm testing distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT. Both the SSD2 and Global methods are the least effective. Global method has poor differentiation between defects and non-defective regions, and SSD2 method displays unclear boundaries of defects. The SSD1 method presents better results than the above methods, but this method still generates unclear boundaries of defects, especially at

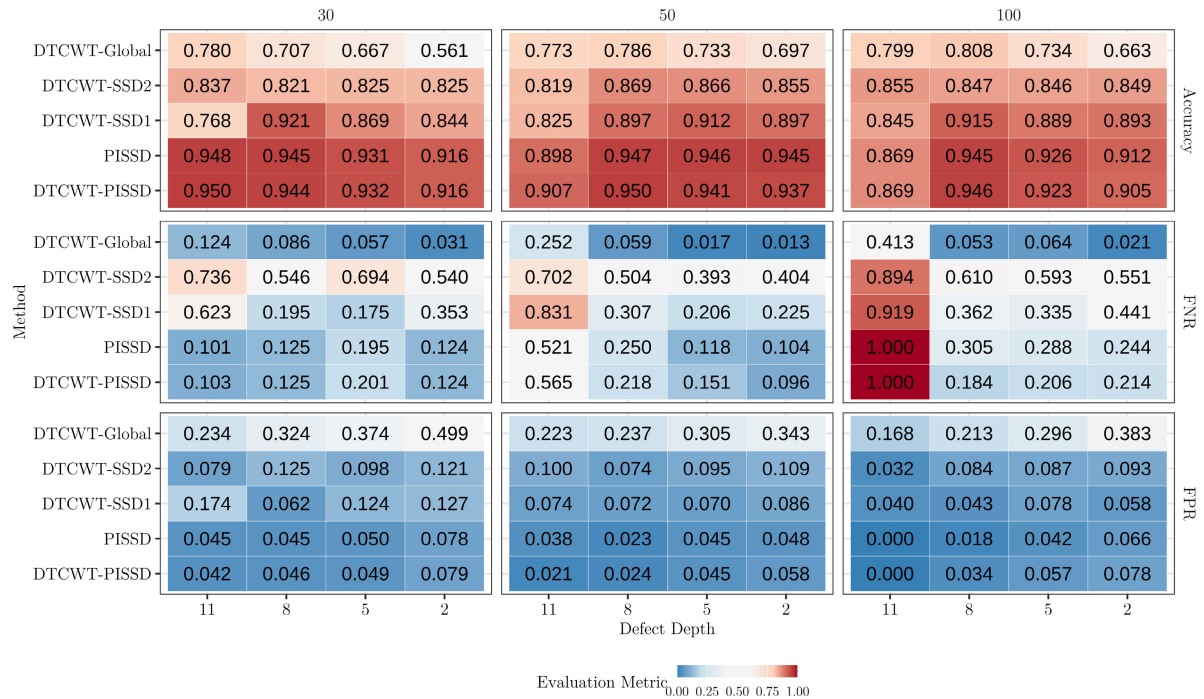




**FIGURE 18.** Proposed image decomposition for multi-depth & multi-size test. The MUT was positioned at a 30 mm distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT.



**FIGURE 19.** Detected defects using different methods for multi-depth & multi-size test on generated raw images. The MUT was positioned at a 30 mm distance from the radar sensor configured with TX3-RX2 transceiver. The image was preprocessed using DTCWT.



**FIGURE 20.** Comprehensive analysis with evaluation metric: Accuracy, FNR, FPR of proposed method DTCWT-PISSD, PISSD and other methods across different defect depth (11 mm, 8 mm, 5 mm, 2 mm) in the multi-depth & multi-size tests. The distance between radar sensor and MUTs ( $z_0$ ) is 30 mm, 50 mm and 100 mm, respectively.

a defect depth of 11 mm. These unclear boundaries of defects occur because at greater depths, the penetration

loss of the signals becomes more severe and degrades the reflected signal in the mmWave system. However,



the PISSD method demonstrates superior performance by clearly identifying defect boundaries at an 11 mm depth as well as at other depths. In summary, the comparison highlights the robustness and effectiveness of the PISSD method in accurately detecting internal defects of varying depths and sizes, outperforming the other evaluated methods, particularly for deeper and smaller defects.

To provide a comprehensive analysis, metrics were used to evaluate image decomposition methods across various depths and sizes of internal defects at three testing distances, as shown in Fig. 20. The results indicate that the proposed automated defect detection algorithms outperform other defect detection algorithms in almost all test metrics across varying defect depths. For instance, at a distance of 30 mm, the PISSD algorithm achieves high accuracy values of 0.948, 0.945, 0.931, and 0.916 for defect depths of 11 mm, 8 mm, 5 mm, and 2 mm, respectively, while the DTCWT-PISSD algorithm records accuracy values of 0.950, 0.944, 0.932, and 0.916 for the same depths. At this 30 mm distance, compared to other methods, the proposed algorithms also achieve the highest performance in FNR and FPR, with values ranging from 0.101 to 0.201 and from 0.042 to 0.079, respectively. These results suggest that the proposed algorithms effectively detect actual defects at various depths due to their high accuracy and low FNR and FPR. The other methods show poor performance, and exhibit imbalances in evaluation metrics, particularly in DTCWT-Global. DTCWT-Global demonstrates high performance in FNR (from 0.031 to 0.124) at a 30 mm testing distance. However, both accuracy and FPR yield low performance, ranging from 0.561 to 0.780 and from 0.234 to 0.499, respectively. Overall, the proposed algorithms demonstrate robustness by optimizing performance across varying defect depths and sizes, reducing false alarms and minimizing the risk of missing critical defects.

## V. LIMITATIONS AND FUTURE WORK

The current study demonstrates the effectiveness of the proposed near-field SAR mmWave imaging system for automated internal defect detection, achieving an average accuracy of 91.7% across various experimental scenarios. However, several limitations remain to be addressed:

- **Testing materials:** The proposed system's performance was tested on 3D-printed materials with controlled defect properties. However, the detection effectiveness of the proposed method in other materials, such as composites, plastic, wood, and concrete, remains to be thoroughly evaluated. Future studies should validate the system's performance across a broader range of materials to ensure its robustness in practical industrial applications.
- **Denoising algorithms:** Denoising algorithm (DTCWT) demonstrated effectiveness in reducing noise within near-field SAR images as shown by the PSNR metric and accuracy in Fig. 8 and Fig. 9, respectively. However, DTCWT may blur defect boundaries and diminish

background clarity. The disturbance patterns may not be accurately captured in image decomposition process, leading to misidentification along the defect boundaries. For example, proposed image decomposition enhanced by DTCWT shows slightly inferior evaluation metrics compared to the one without DTCWT in certain cases in Fig. 12, Fig. 16, and Fig. 20. Therefore, future denoising algorithms should be further optimized to eliminate noise in near-field SAR images while preserving defect boundaries in subsequent studies.

- **Enhanced defect detection:** The proposed framework for automated internal defect localization algorithm demonstrates promising results. In future research, implementation of multi-input multi-output (MIMO) antenna arrays could improve detection accuracy by capturing reflected signals from multiple angles, thereby providing more accurate characterization of defect shapes and sizes. Furthermore, beamforming techniques can be employed to enhance signal penetration depth, particularly for materials with high permittivity or deeply located defects, thereby increasing the detection capabilities of the MMIC-based mmWave NDT testing system while maintaining its compact form factor.

These limitations present opportunities for future research to improve the speed, versatility, and accuracy of the near-field SAR mmWave NDT system, making it more adaptable to real-world industrial applications.

## VI. CONCLUSION

This paper presents an automated, fast, and low-cost NDT system using mmWave near-field imaging. We have developed a robust scanning system capable of characterizing the internal structures of test objects by producing high-resolution SAR images. The quality of these images has been enhanced through a denoising algorithm that mitigates noise in the near-field context. The defect detection algorithm has automatically and accurately identified defects on the generated images despite the non-smooth and dynamic background patterns in near-field SAR images. This automated algorithm expedites the detection and reduces the operator-induced false identification. We have demonstrated the efficacy of the proposed system through detecting internal defects across variations in shape, material permittivity, depth, and size. Comparative analysis underscores the superior performance of our automated defect detection approach in NDT relative to existing methods. The compact size of the radar sensor enables this framework to easily integrate with larger scanning systems, facilitating defect detection in objects of any size. Moreover, the automated defect detection is essential for software development, which can be a potential module within the sensing system. This integration enhances efficiency of rapid internal health inspections. Overall, this advancement in the fast, low-cost, and automated framework for internal defect detection paves the way for more efficient, accurate, and in-situ applications in structural health monitoring.

## REFERENCES

- [1] P. J. Shull, *Nondestructive Evaluation: Theory, Techniques, and Applications*. Boca Raton, FL, USA: CRC Press, 2002.
- [2] A. Ghavamian, F. Mustapha, B. T. H. T. Baharudin, and N. Yidris, "Detection, localisation and assessment of defects in pipes using guided wave techniques: A review," *Sensors*, vol. 18, no. 12, p. 4470, Dec. 2018.
- [3] J. Bustillo, H. Achdjian, A. Arciniegas, and L. Blanc, "Simultaneous determination of wave velocity and thickness on overlapped signals using forward backward algorithm," *NDT E Int.*, vol. 86, pp. 100–105, Mar. 2017.
- [4] Z. Zhenwei, X. Jing, J. Rui, W. Yinghong, G. Hao, H. Siyi, Z. Cunlin, and Z. Yuejin, "Terahertz non-destructive testing and imaging of high-voltage cables," *Frontiers Phys.*, vol. 10, Apr. 2022, Art. no. 893145.
- [5] R. Schurch, S. M. Rowland, R. S. Bradley, and P. J. Withers, "Imaging and analysis techniques for electrical trees using X-ray computed tomography," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 21, no. 1, pp. 53–63, Feb. 2014.
- [6] C. Zhang, "Characteristics and applications of non-destructive testing technology in bridge engineering," *Highlights Sci., Eng. Technol.*, vol. 75, pp. 221–227, Dec. 2023.
- [7] Z. Li, Z. Meng, J. Liu, C. Soutis, and A. Gibson, "Machine learning-enabled thickness estimation of thin coatings on carbon fibre composites using microwaves," *Meas. Sci. Technol.*, vol. 34, no. 12, Dec. 2023, Art. no. 124003.
- [8] Y. Fang, Z. Chen, X. Yang, R. Wang, Y. Li, and S. Xie, "Visualization and quantitative evaluation of delamination defects in GFRPs via sparse millimeter-wave imaging and image processing," *NDT E Int.*, vol. 141, Jan. 2024, Art. no. 102975.
- [9] I. Jaeger, L. Zhang, J. Stiens, H. Sahli, and R. Vounckx, "Millimeter wave inspection of concealed objects," *Microw. Opt. Technol. Lett.*, vol. 49, no. 11, pp. 2733–2737, Nov. 2007.
- [10] D. M. Sheen, D. L. McMakin, and T. E. Hall, "Three-dimensional millimeter-wave imaging for concealed weapon detection," *IEEE Trans. Microw. Theory Techn.*, vol. 49, no. 9, pp. 1581–1592, Sep. 2001.
- [11] M. Becquaert, E. Cristofani, H. Van Luong, M. Vandewal, J. Stiens, and N. Deligiannis, "Compressed sensing mm-wave SAR for non-destructive testing applications using multiple weighted side information," *Sensors*, vol. 18, no. 6, p. 1761, May 2018.
- [12] S. Kharkovsky and R. Zoughi, "Microwave and millimeter wave nondestructive testing and evaluation—Overview and recent advances," *IEEE Instrum. Meas. Mag.*, vol. 10, no. 2, pp. 26–38, Apr. 2007.
- [13] M. E. Yanik, D. Wang, and M. Torlak, "Development and demonstration of MIMO-SAR mmWave imaging testbeds," *IEEE Access*, vol. 8, pp. 126019–126038, 2020.
- [14] Y. Huang, Y. Shen, and J. Wang, "From terahertz imaging to terahertz wireless communications," *Engineering*, vol. 22, pp. 106–124, Mar. 2023.
- [15] C. Schwäbig, S. Wang, and S. Gütermann, "Development of a millimetre wave based SAR real-time imaging system for three-dimensional non-destructive testing," *Technisches Messen*, vol. 88, nos. 7–8, pp. 488–497, Aug. 2021.
- [16] X. Zhang, J. Liang, N. Wang, T. Chang, Q. Guo, and H.-L. Cui, "Broad-band millimeter-wave imaging radar-based 3-D holographic reconstruction for nondestructive testing," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 3, pp. 1074–1085, Mar. 2020.
- [17] A. Batra, J. Barowski, D. Damyanov, M. Wiemeler, I. Rolfes, T. Schultze, J. C. Balzer, D. Göhringer, and T. Kaiser, "Short-range SAR imaging from GHz to THz waves," *IEEE J. Microw.*, vol. 1, no. 2, pp. 574–585, Apr. 2021.
- [18] M. Caris, S. Stanko, A. Wahlen, R. Sommer, J. Wilcke, N. Pohl, A. Leuther, and A. Tessmann, "Very high resolution radar at 300 GHz," in *Proc. 44th Eur. Microw. Conf.*, Oct. 2014, pp. 1797–1799.
- [19] D. Hou, J. Chen, P. Yan, and W. Hong, "A 270 GHz×9 multiplier chain MMIC with on-chip dielectric-resonator antenna," *IEEE Trans. THz Sci. Technol.*, vol. 8, no. 2, pp. 224–230, Mar. 2018.
- [20] S. Thomas, C. Bredendiek, T. Jaeschke, F. Vogelsang, and N. Pohl, "A compact, energy-efficient 240 GHz FMCW radar sensor with high modulation bandwidth," in *Proc. German Microw. Conf. (GeMiC)*, Mar. 2016, pp. 397–400.
- [21] Y. Shang, H. Yu, H. Fu, and W. M. Lim, "A 239–281 GHz CMOS receiver with on-chip circular-polarized substrate integrated waveguide antenna for sub-terahertz imaging," *IEEE Trans. THz Sci. Technol.*, vol. 4, no. 6, pp. 686–695, Nov. 2014.
- [22] S. Sun, A. P. Petropulu, and H. V. Poor, "MIMO radar for advanced driver-assistance systems and autonomous driving: Advantages and challenges," *IEEE Signal Process. Mag.*, vol. 37, no. 4, pp. 98–117, Jul. 2020.
- [23] M. Schneider, "Automotive radar-status and trends," in *Proc. German Microwave Conf.*, 2005, pp. 144–147.
- [24] S. Gishkori, L. Daniel, M. Gashinova, and B. Mulgrew, "Imaging for a forward scanning automotive synthetic aperture radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 3, pp. 1420–1434, Jun. 2019.
- [25] X. Mao, "Spherical geometry algorithm for space-borne synthetic aperture radar imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5217215. [Online]. Available: <https://ieeexplore.ieee.org/document/10595442>
- [26] C. Qu, "Research on near field signal behaviour of localizer antenna system," *J. Phys., Conf.*, vol. 2449, no. 1, Mar. 2023, Art. no. 012046.
- [27] A. Kolkiran and G. S. Agarwal, "Resolution of objects within subwavelength range by using the near field of a dipole," *Opt. Lett.*, vol. 37, no. 12, pp. 2313–2315, 2012.
- [28] T.-H. Pham, K.-H. Kim, and I.-P. Hong, "A study on millimeter wave SAR imaging for non-destructive testing of rebar in reinforced concrete," *Sensors*, vol. 22, no. 20, p. 8030, Oct. 2022.
- [29] M. ELsaadouny, J. Barowski, J. Jebramcik, and I. Rolfes, "Millimeter wave SAR imaging for the non-destructive testing of 3D-printed samples," in *Proc. Int. Conf. Electromagn. Adv. Appl. (ICEAA)*, Sep. 2019, pp. 1283–1285.
- [30] H. Dai, Y. He, W. Luo, F. Zhang, and S. Pan, "Near-field millimetre wave SAR imaging for non-destructive testing," in *Proc. IET Int. Radar Conf. (IRC)*, 2023, pp. 4128–4131.
- [31] P. Singh, M. Diwakar, A. Shankar, R. Shree, and M. Kumar, "A review on SAR image and its despeckling," *Arch. Comput. Methods Eng.*, vol. 28, no. 7, pp. 4633–4653, Dec. 2021.
- [32] Y. Kim, S. Boo, G. Kim, N. Kim, and B. Lee, "Wireless power transfer efficiency formula applicable in near and far fields," *J. Electromagn. Eng. Sci.*, vol. 19, no. 4, pp. 239–244, Oct. 2019.
- [33] F. Rosu, A. Anghel, S. Ciochina, R. Cacoveanu, and M. Datcu, "Near-range multipath mitigation methodology for multistatic SAR applications using matched-adaptive filters," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3204–3214, 2022.
- [34] T. Zeng, W. Zhang, X. Zhan, X. Xu, Z. Liu, B. Wang, and X. Zhang, "A novel multimodal fusion framework based on point cloud registration for near-field 3D SAR perception," *Remote Sens.*, vol. 16, no. 6, p. 952, Mar. 2024.
- [35] L. Li, X. Zhang, Y. Zhou, L. Pu, J. Shi, and S. Wei, "Region adaptive morphological reconstruction fuzzy C-means for near-field 3-D SAR image target extraction," *Digit. Signal Process.*, vol. 113, Jun. 2021, Art. no. 103036.
- [36] M. E. Yanik and M. Torlak, "Near-field MIMO-SAR millimeter-wave imaging with sparsely sampled aperture data," *IEEE Access*, vol. 7, pp. 31801–31819, 2019.
- [37] P. M. Bhatt, R. K. Malhan, P. Rajendran, B. C. Shah, S. Thakar, Y. J. Yoon, and S. K. Gupta, "Image-based surface defect detection using deep learning: A review," *J. Comput. Inf. Sci. Eng.*, vol. 21, no. 4, Aug. 2021, Art. no. 040801.
- [38] B. Szymanik, G. Psuj, M. Hashemi, and P. Lopato, "Detection and identification of defects in 3D-printed dielectric structures via thermographic inspection and deep neural networks," *Materials*, vol. 14, no. 15, p. 4168, Jul. 2021.
- [39] B. Szymanik, "An evaluation of 3D-printed materials' structural properties using active infrared thermography and deep neural networks trained on the numerical data," *Materials*, vol. 15, no. 10, p. 3727, May 2022.
- [40] N. H. M. M. Shrifan, G. N. Jawad, N. A. M. Isa, and M. F. Akbar, "Microwave nondestructive testing for defect detection in composites based on K-means clustering algorithm," *IEEE Access*, vol. 9, pp. 4820–4828, 2021.
- [41] X. Yang, Y. Fang, R. Wang, Y. Li, and Z. Chen, "Visual quantitative detection of delamination defects in GFRP via microwave," *Sensors*, vol. 23, no. 14, p. 6386, Jul. 2023.
- [42] S. D. Silverstein and Y. Zheng, "Near-field inverse coherent imaging problems: Solutions, simulations and applications," in *Proc. 37th Asilomar Conf. Signals, Syst. Comput.*, vol. 1, 2003, pp. 1193–1197.
- [43] H. Weyl, "Ausbreitung elektromagnetischer Wellen über einem ebenen Leiter," *Annalen der Physik*, vol. 365, no. 21, pp. 481–500, 1919.
- [44] A. R. Barket, W. Hu, B. Wang, W. Shahzad, and J. S. Malik, "Selection criteria of image reconstruction algorithms for terahertz short-range imaging applications," *Opt. Exp.*, vol. 30, no. 13, pp. 23398–23416, 2022.

- [45] G. Wang, F. Qi, Z. Liu, C. Liu, C. Xing, and W. Ning, "Comparison between back projection algorithm and range migration algorithm in terahertz imaging," *IEEE Access*, vol. 8, pp. 18772–18777, 2020.
- [46] B. Kumar, R. K. Ranjan, and A. Husain, "A multi-objective enhanced fruit fly optimization (MO-EFOA) framework for despeckling SAR images using DTCWT based local adaptive thresholding," *Int. J. Remote Sens.*, vol. 42, no. 14, pp. 5493–5514, Jul. 2021.
- [47] I. Selesnick, R. G. Baraniuk, and N. Kingsbury, "The dual-tree complex wavelet transform," *IEEE signal Process. Mag.*, vol. 22, no. 6, pp. 123–151, Nov. 2005.
- [48] F. Luisier, T. Blu, and M. Unser, "A new SURE approach to image denoising: Interscale orthonormal wavelet thresholding," *IEEE Trans. Image Process.*, vol. 16, no. 3, pp. 593–606, Mar. 2007.
- [49] J. Antkowiak, T. J. Baina, F. V. Baroncini, N. Chateau, F. FranceTelecom, A. C. F. Pessoa, F. S. Colonnese, I. L. Contin, J. Caviedes, and F. Philips, "Final report from the video quality experts group on the validation of objective models of video quality assessment March 2000," Video Qual. Experts Group (VQEG), Int. Telecommun. Union (ITU), Telecommun. Standardization Sector (ITU-T), Tech. Rep. COM 9-80-E, vol. 10, 2000. [Online]. Available: [https://www.itu.int/ITU-T/studygroups/com09/docs/tutorial\\_opavc.pdf](https://www.itu.int/ITU-T/studygroups/com09/docs/tutorial_opavc.pdf)
- [50] H. Yan, K. Paynabar, and J. Shi, "Anomaly detection in images with smooth background via smooth-sparse decomposition," *Technometrics*, vol. 59, no. 1, pp. 102–114, Jan. 2017.
- [51] L. Xiao, Y. Li, and D. Ruppert, "Fast bivariate  $P$ -splines: The sandwich smoother," *J. Roy. Stat. Soc. B, Stat. Methodol.*, vol. 75, no. 3, pp. 577–599, Jun. 2013.
- [52] M. T. Güneşer and F. Atasoy, "Extracting complex permittivity of materials by Gaussian process regression using the transmission parameter at sub-THz," *J. Electron. Mater.*, vol. 49, no. 1, pp. 681–688, Jan. 2020.
- [53] A. Och, P. A. Hözl, S. Schuster, S. Scheibelhofer, D. Zankl, V. Pathuri-Bhuvana, and R. Weigel, "High-resolution millimeter-wave tomography system for nondestructive testing of low-permittivity materials," *IEEE Trans. Microw. Theory Techn.*, vol. 69, no. 1, pp. 1105–1113, Jan. 2021.
- [54] I. Kuzmanić, I. Vujović, M. Petković, and J. Šoda, "Influence of 3D printing properties on relative dielectric constant in PLA and ABS materials," *Prog. Additive Manuf.*, vol. 8, no. 4, pp. 703–710, Aug. 2023.



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nondestructive defect detection in materials.

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